Bayesian Surface Habitability Mapping of Titan

from the Late Heavy Bombardment to the Solar Red-Giant Phase

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Planetary Science and Life in the Universe

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All code, processed data products, and static habitability maps are openly archived at `https://github.com/chrimblechrumble/pslu-p3`.

1 Introduction

1.1 The Search for Habitable Environments Beyond Earth

The question of whether life exists elsewhere in the universe has undergone a profound transformation over the past three decades, shifting from the domain of philosophical speculation to empirical scientific investigation. This change has been driven by two largely independent but mutually reinforcing developments: the systematic discovery that life on Earth thrives in conditions previously considered detrimental to biology, and the growing recognition through robotic spacecraft exploration that chemically and thermodynamically suitable environments for life are far more widely distributed throughout the solar system than twentieth-century intuitions suggested.

The progressive characterisation of terrestrial extremophile communities has expanded the limits of conditions considered compatible with life. Microorganisms have been cultured from deep-sea hydrothermal vents at temperatures exceeding 121°C, from hypersaline brines with water activities as low as $a_w = 0.60$, from the anoxic pore spaces of deep crustal rocks where electrons liberated by water–rock reactions provide metabolic energy in the total absence of sunlight, and from sub-glacial Antarctic lakes that have been isolated from the atmosphere for millions of years (Cavicchioli, 2002; Rothschild and Mancinelli, 2001). Each such discovery has moved the habitable niche boundaries outwards, making it progressively more difficult to argue that any particular planetary environment is categorically uninhabitable without specific evidence that the relevant physical and chemical conditions lie entirely outside the known boundaries of life.

In parallel, the outer solar system has revealed a far richer array of liquid environments than the classical habitable zone framework would predict. The Europa detection from the Galileo mission (Pappalardo et al., 1999) and the subsequent discovery of active cryovolcanism and ocean chemistry at Enceladus (Waite et al., 2017) established that tidal heating can maintain liquid water oceans beneath ice shells at distances far beyond any reasonable definition of the water-based habitable zone (Kasting et al., 1993). These discoveries shifted the conceptual framework of solar system habitability from a two-dimensional picture (Earth-like planets in a narrow orbital band) to a three-dimensional one encompassing subsurface environments throughout the outer solar system.

Within this broader astrobiological context, Saturn's moon Titan occupies a distinctive and arguably singular position. It is the only moon in the solar system possessing a substantial atmosphere — one that is denser at the surface than Earth's and dominated by molecular nitrogen, with a structurally complex photochemistry driven by the ultraviolet flux of the distant Sun. It is the only known extraterrestrial body to host stable surface liquids (Stofan et al., 2007), completing a full precipitation-evaporation-lake cycle that is the structural analogue of the terrestrial hydrological cycle. It may have accumulated the most complex organic chemical inventory in the solar system outside Earth itself (Lorenz et al., 2008). And at depth, it contains a water–ammonia ocean whose thermodynamic conditions are increasingly well-constrained by tidal measurements and geophysical modelling. The combination of these properties makes Titan not merely a potentially habitable world but a natural laboratory for prebiotic chemistry operating at planetary scale and across a geological timescale of ~4.5 Gyr.

1.2 The Physical Environment of Titan

Titan's dense nitrogen–methane atmosphere (~94% N_2, ~5.6% CH_4; surface pressure ~1.5 bar, surface temperature ~94 K; Fulchignoni et al. 2005; Niemann et al. 2005) drives a methane cycle analogous to Earth's water cycle, with rain, rivers, and polar hydrocarbon lakes. Ultraviolet photolysis of the N_2/CH_4 mixture continuously produces tholins at $\sim 5 \times 10^{-14}\,\mathrm{g\,cm^{-2}\,s^{-1}}$ (Lavvas et al., 2011), building a global organic sediment layer estimated at 1–10 km depth over solar system history (Lorenz et al., 2008).

The surface is geologically diverse: equatorial organic dune fields (~17% of the SAR-imaged area), plains (~65%), and a north polar sea system — Kraken Mare, Ligeia Mare, and Punga Mare — containing an estimated 70,000 km^3 of liquid hydrocarbons (Malaska et al., 2025; Mastrogiuseppe et al., 2019). The Lopes et al. (2019) global geomorphological map, derived from the full Cassini SAR dataset, provides the terrain classification underpinning the organic abundance feature used in this thesis.

The interior harbours a global water–ammonia subsurface ocean at ~55–80 km depth, inferred from the tidal Love number $k_2 = 0.589 \pm 0.150$ (Iess et al., 2012). This ocean provides a potential habitat in its own right and, at past epochs, a source of liquid water during impact-melt events where ocean-derived chemistry contacts the surface organic inventory (Tobie et al., 2005).

1.3 Chemical Habitability Scenarios

Four habitability pathways have been proposed for Titan's surface.

1. The *acetylenotrophy hypothesis* (McKay and Smith, 2005) identifies

$$C_2H_2 + 3\,H_2 \longrightarrow 2\,CH_4 \qquad (\Delta G = -334\,kJ\,mol^{-1}) \tag{1.1}$$

   as a viable energy source at the present epoch, with acetylene and H_2 supplied by atmospheric photolysis; the energetics comfortably exceed the metabolic threshold of terrestrial methanogens even under updated thermochemical estimates (Yanez et al., 2024).

2. The *azotosome hypothesis* (Stevenson et al., 2015) proposes that acrylonitrile (detected by ALMA; Palmer et al. 2017) spontaneously forms stable membrane-like vesicles in liquid methane, providing a plausible route to cellular compartmentalisation without aqueous chemistry.

3. *Post-impact melt chemistry* offers a water-based pathway: large impacts produce water–ammonia–organic melt ponds persisting for 10^3–10^4 yr during which Strecker synthesis yields amino acids from surface aldehydes, ammonia, and HCN (Madan et al., 2026; Neish et al., 2010; wiki:Strecker, 2026); this is the primary astrobiological rationale for the *Dragonfly* landing site at Selk crater.
4. Finally, the *far-future habitability window* (Lorenz et al., 1997) arises when red-giant solar luminosity raises Titan's surface above the water–ammonia eutectic (176 K) for an estimated 200–500 Myr, creating a global ocean over 16 Gyr of accumulated organics.

This thesis provides a quantitative characterisation of all four scenarios within a unified Bayesian framework across geological time.

1.4 Scope and Research Questions

Several preceding studies have attempted to place planetary habitability on a quantitative footing. The *Earth Similarity Index* (Schulze-Makuch et al., 2011) scores planetary bodies against Earth on temperature, density, escape velocity, and surface chemistry, but produces a single scalar per body and cannot represent spatial or temporal variation. Hendrix et al. (2019) reviewed the habitability potential of icy ocean worlds using qualitative multi-criteria matrices, providing rich scientific context but no spatial resolution or probabilistic framework. Logistic regression and machine-learning habitability classifiers (e.g. Spiegel and Turner 2012) have been applied to exoplanet populations but require labelled training data that does not exist for Titan. The Bayesian approach adopted here differs from all of these: it encodes scientific prior knowledge as probability distributions, updates them with observational data via conjugate updating, and returns credible intervals rather than point estimates — making uncertainty explicit rather than implicit. The choice of a Beta-conjugate (Beta-Binomial) model is motivated by the bounded $[0, 1]$ domain of habitability probability and by the analytical tractability that permits simultaneous evaluation at all 6.49 million grid pixels without numerical integration or MCMC. A Gaussian process or multivariate logistic regression could in principle capture feature correlations more richly, but would require either strong covariance priors or prohibitively dense labelled sampling to estimate the correlation structure from Cassini data alone. The Beta-conjugate approach is preferred as the most tractable framework that preserves physical interpretability at every pixel.

Previous habitability assessments of Titan have shared four important limitations.

1. They are predominantly qualitative or point-specific, lacking global spatial resolution.
2. They treat the present epoch as the only relevant state, ignoring both the LHB prebiotic-chemistry window and the far-future red-giant window.
3. They do not integrate the full suite of Cassini data modalities into a unified probabilistic framework.

4. They lack the reproducibility afforded by an openly archived analysis pipeline.

The closest methodological precedent is Affholder et al. (2021), who applied Bayesian statistical inference to Cassini INMS plume data from Enceladus, comparing the observed methane and hydrogen escape rates against models of abiotic serpentinisation and biotic methanogenesis. Their finding — that the observed methane escape rate is (with a minimal number of assumptions) explained by biological methanogenesis if the prior probability of life is sufficient — represents a comparable Bayesian biosignature assessment for a solar system body. This thesis is directly inspired by their approach, extended from a single-point plume observation to a full-globe, multi-feature, multi-epoch spatial mapping.

The specific research questions addressed are:

RQ1 *What is the spatial distribution of surface habitability on present-day Titan, and which terrain types, geological features, and atmospheric zones produce the highest scores? Are the data-derived feature importances consistent with the physically motivated prior weights?*

RQ2 *How did habitability differ during the Late Heavy Bombardment, when impact-melt ponds provided transient liquid water? Which impact structures were the most astrobiologically significant, and what are the implications for prebiotic chemistry?*

RQ3 *How does the surface habitability evolve over geologic time from the LHB to the red-giant future? Is Titan potentially the most habitable body in the solar system during the red-giant ocean window?*

RQ4 *Does the derived habitability map independently validate the Dragonfly landing site at Selk crater, and which instruments are best positioned to test the model's predictions? See Appendix G for more information on the Cassini–Huygens and Dragonfly missions.*

Chapter 2 presents the data sources, feature extraction, Bayesian inference model, and temporal methodology. Chapter 3 presents and interprets the habitability maps at each epoch, the feature importances, the temporal evolution, and the detailed Selk crater analysis. Chapter 4 discusses the *Dragonfly* mission implications, biosignature framework, limitations, and future work.

2 Methods

Note: Acronyms and notation are defined in Appendix E

2.1 Overview of the Analytical Approach

The framework proceeds in the following stages:

1. A one-off data-acquisition stage to gather relevant datasets from varied raw and derived sources. Details are presented in Appendix E.3
2. A data preprocessing stage: e.g. some map data needs converting into a canonical mapping projection so that different features can be computed/extracted from the same Titan location. The canonical format used is
   (a) Projection: SimpleCylindrical (equirectangular)
   (b) Titan sphere: R=2,575,000 m
   (c) Longitude: West-positive, $0° \rightarrow 360°$, (matches all USGS raster products)
   (d) Latitude: North-up, $+90° \rightarrow -90°$
3. Extraction of 8 habitability-proxy features from the data, details in sections 2.2, 2.3.
4. Bayesian conjugate updating to produce a posterior habitability probability map at each of five named geological 'anchor' epochs, (*Past*, *Lake Formation*, *Present*, *Near Future*, *Future*). Details are presented in 2.4
5. Outputs: Graphs and ranking of named surface sites by posterior probability for each epoch. Details in 2.5

Figure 2.1 provides a schematic overview of the five-stage analytical pipeline from data acquisition to visualisation.

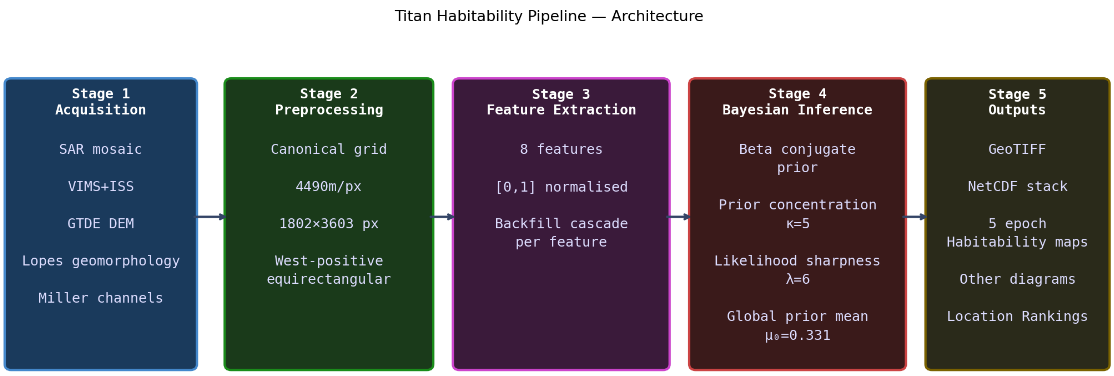

Figure 2.1: Schematic overview of the five-stage Titan habitability analysis pipeline. All eight habitability-proxy features are extracted in Stage 3 and combined in the Bayesian posterior update of Stage 4 to produce $P(H \mid \mathbf{f})$ maps at each of the five epochs.

The Bayesian approach taken in this thesis requires explanation and follows next.

2.1.1 The Bayesian Approach: Motivation and Structure

The central methodological challenge in quantifying Titan's surface habitability is that no single Cassini instrument directly measures habitability. Each instrument provides a partial and indirect proxy: radar backscatter reveals surface composition and liquid surfaces; near-infrared spectroscopy constrains organic content; thermal emission maps the temperature field that drives the methane cycle; altimetry constrains topography and hence organic accumulation patterns; and the geomorphological classification synthesises the collective information from all instruments into terrain types whose habitability implications can be grounded in laboratory and theoretical work.

The approach taken here is to combine these partial proxies through a Bayesian inference framework that produces, at each grid cell, a posterior probability of habitability $P(H \mid \mathbf{f})$ given the vector of observed features $\mathbf{f} = (f_1, \ldots, f_8)$. This is structurally analogous to the approach of Affholder et al. (2021), who used Bayesian methods to evaluate whether the observed methane flux in Enceladus's plumes is more consistent with abiotic serpentinisation or with biological methanogenesis. The key extensions here are spatial resolution (a 6.49-million-pixel global map rather than a single-point estimate), the number of independent observational inputs (eight features rather than one), and the temporal scope (five epochs spanning geological time rather than a single snapshot).

The structure of the inference is:

1. encode prior knowledge about Titan's habitability into a probability distribution over the latent variable H;

2. update that distribution using each observed feature f_i as evidence;

3. report the resulting posterior probability $P(H \mid \mathbf{f})$ as the habitability estimate at each pixel.

The Bayesian inference and subsequernt analysis is applied at 4490 m/px resolution on a canonical equirectangular grid covering the full globe (1802 pixels in latitude $\times$ 3603 pixels in longitude), using a west-positive longitude convention consistent with all Cassini raster products. This resolution is chosen to match the coarsest primary dataset (the GTDR altimetry mosaic at $\sim$22 km per pixel) whilst resolving lake margins and fluvial features at $\sim$45 km scale. All eight input datasets are reprojected to this common grid before feature extraction.

2.2 Observational Data Sources

Eight Cassini data products are used, summarised in Table 2.1. Processing details are in Appendix E.

Table 2.1: Cassini data sources used in the habitability framework.

Dataset	Coverage	Habitability feature
Cassini SAR mosaic	∼67% surface	f_1 liquid hydrocarbon; f_7 geodiversity
VIMS 1.59/1.27 µm	∼50% surface	f_2 organic abundance
CIRS thermal mosaic	Global (∼10°)	f_4 methane cycle; f_5 surf–atm
GTDE/GTDR DEM	∼90% (interp.)	f_6 topographic complexity
Lopes (2019) geomorphology map	∼90% SAR	f_2 organic; f_7 geodiversity
Miller (2021) channel map	SAR coverage	f_5 surface–atmosphere
Hedgepeth (2020) craters	SAR coverage	f_7 geodiversity; past epoch
Tidal k_2 (global)	Global scalar	f_8 subsurface ocean

2.3 Feature Extraction

Eight habitability-proxy features f_1–f_8 are extracted from the data sources above, each normalised to $[0, 1]$ via robust percentile clipping (2nd–98th percentile; Eq. 2.1). Table 2.2 summarises the features, their prior weights, prior means, and scientific basis. Full engineering details — data processing, normalisation choices, seam-correction rationale for f_2, and backfill procedures — are in Appendix F.

$$\hat{f}_i = \mathrm{clip}\left(\frac{f_i - f_{i,\mathrm{p2}}}{f_{i,\mathrm{p98}} - f_{i,\mathrm{p2}}},\ 0,\ 1\right) \tag{2.1}$$

Table 2.2: The eight habitability-proxy features. Weight w_i and prior mean μ_i are the present-epoch values used in the Bayesian update (Eq. 2.8). $\mu_0 = \sum_i w_i \mu_i = 0.331$.

f_i	Name	w_i	μ_i	Scientific rationale
f_1	Liquid hydrocarbon availability	0.25	0.02	Direct solvent for non-aqueous biochemistry (McKay and Smith, 2005)
f_2	Organic abundance	0.20	0.70	Tholin substrate for prebiotic chemistry (Meyer-Dombard et al., 2025)
f_3	Chemical energy (acetylene)	0.20	0.35	Acetylenotrophy energy proxy (McKay and Smith, 2005; Yanez et al., 2024)
f_4	Methane cycle intensity	0.15	0.40	Solvent transport and chemical concentration cycling
f_5	Surface–atm. interaction	0.08	0.35	Atmospheric delivery of organics and energy substrates
f_6	Topographic complexity	0.06	0.25	Microenvironment diversity and chemical gradient maintenance
f_7	Geomorphological diversity	0.04	0.30	Terrain heterogeneity proxy for ecological niche richness
f_8	Subsurface ocean proximity	0.02	0.03	Potential water-chemistry conduit; SAR-bright annuli (Iess et al., 2012)

Figure F.1 shows all eight feature maps at the present epoch; Figure F.2 shows how each feature is modulated across geological time.

2.4 Bayesian Inference Model

2.4.1 Formulation: Latent Variables, Conjugacy, and the Beta-Binomial Model

The latent variable. In this framework, habitability H is treated as a *latent variable* — a quantity that cannot be directly observed by any Cassini instrument but whose value is inferred from the combination of observational proxies $\mathbf{f} = (f_1, \ldots, f_8)$. A latent variable (from the Latin *latere*: to be hidden) influences observable outcomes but is not itself directly measurable. Here, H is not a binary yes-or-no property but a continuous probability in $[0, 1]$: the value $H = p$ at a given pixel means that pixel has probability p of satisfying the relevant habitability conditions given all available evidence. Each feature f_i provides partial, indirect evidence about the value of H.

Why a Beta distribution? The Beta distribution $\mathrm{Beta}(\alpha, \beta)$ is defined on $[0, 1]$ and characterised by two positive shape parameters α and β. It is unimodal when both $\alpha > 1$ and $\beta > 1$, with mean $\alpha/(\alpha + \beta)$ and variance that decreases as $\alpha + \beta$ grows. Because it is bounded on $[0, 1]$ and can represent any unimodal belief over a probability, it is the natural prior for the latent habitability probability H.

Conjugacy explained. A prior distribution is *conjugate* to a likelihood function if the posterior (prior updated by data) has the same mathematical form as the prior, differing only in its parameter values. The Beta distribution is conjugate to the Binomial likelihood: if the prior on a success probability p is $\mathrm{Beta}(\alpha_0, \beta_0)$ and we observe k successes in n Bernoulli trials, the posterior is $\mathrm{Beta}(\alpha_0 + k,\ \beta_0 + (n - k))$. This analytical tractability makes the update feasible at 6.49 million pixels simultaneously with no numerical integration required.

The Beta-Binomial model. Each feature $f_i \in [0, 1]$ is treated as the fractional outcome of λw_i virtual Bernoulli trials: fraction f_i of those trials are "successes" (evidence for habitability) and $(1 - f_i)$ are "failures" (evidence against). High feature values contribute more positive pseudo-observations and shift the posterior towards higher habitability; low values do the opposite. The prior contributes κ virtual observations and the data from all eight features contribute a further $\lambda \sum_i w_i = \lambda$ observations. Figure 2.3 visualises the updating process for three representative sites (Kraken southern shore, Belet dune field, Selk crater), showing how each successive feature narrows and displaces the Beta distribution from the prior to the final posterior. The figure also shows the 95% HDI for eight key present-epoch sites.

Figure 2.2 illustrates this update graphically for four representative sites. The left panel shows the prior $\mathrm{Beta}(\alpha_0, \beta_0)$, which is weakly informative ($\sigma \approx 0.18$) and centred at $\mu_0 = 0.331$. The right panel shows the posteriors after incorporating the respective Cassini feature values: the Kraken Mare shoreline and Ligeia shoreline shift rightward (high liquid, methane-cycle, and geodiversity scores), while the Belet dune sea shifts only modestly (high organic but no liquid) and Selk crater shifts leftward (below-average methane cycle and surface–atmosphere scores despite exceptional geodiversity).

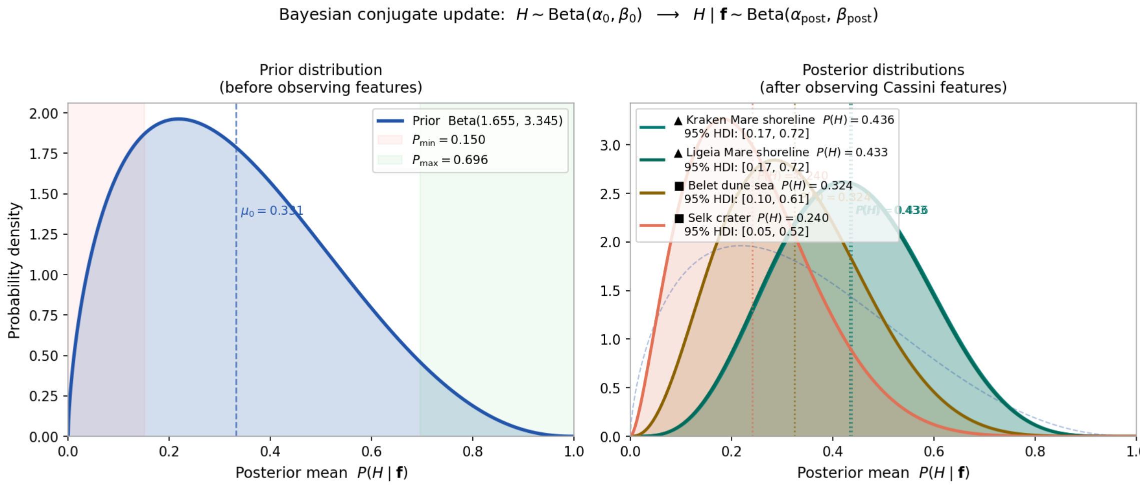

Figure 2.2: Beta-conjugate Bayesian update for four representative Titan surface environments. *Left:* the prior distribution $H \sim \text{Beta}(1.655, 3.345)$ with mean $\mu_0 = 0.331$. Shaded bands mark the hard bounds $[P_{\min}, P_{\max}] = [0.150, 0.696]$. *Right:* posterior distributions after incorporating the Cassini feature vector $\mathbf{f}$ for each site (parameters $\kappa = 5$, $\lambda = 6$; present epoch). The update sharpens and shifts each distribution in proportion to how far the site's weighted feature sum $\sum_i w_i f_i$ departs from the prior mean. Polar lake shores (dark cyan/teal) shift markedly right; the equatorial crater (dark orange-red) shifts left; the dune sea (dark amber) remains near the prior. 95% HDI values are given in the legend.

2.4.2 Prior Distribution

The prior distribution over habitability at any pixel is:

$$H \sim \text{Beta}(\alpha_0, \beta_0), \qquad \alpha_0 = \mu_0 \kappa, \quad \beta_0 = (1 - \mu_0)\kappa \tag{2.2}$$

where μ_0 is the global prior mean habitability and κ is the concentration parameter. The parameter α_0 counts prior pseudo-successes (evidence for habitability) and β_0 counts prior pseudo-failures. The global prior mean is the weighted average of the eight feature prior means:

$$\mu_0 = \sum_{i=1}^{8} w_i \mu_i = 0.331 \tag{2.3}$$

This represents our best estimate that an average Titan pixel has ∼33% probability of habitability before examining the Cassini data. The value reflects the combination of a large organic inventory ($w_2\mu_2 = 0.14$, the dominant term) tempered by limited liquid coverage ($w_1\mu_1 = 0.005$, the smallest term). Full prior values are listed in Appendix B.

The concentration parameter $\kappa = 5$ controls how strongly the prior resists updating: a larger κ requires more data to shift the posterior away from μ_0. The value $\kappa = 5$ was chosen to place the prior and data on approximately equal footing ($\kappa/(\kappa + \lambda) = 5/11 \approx 0.45$), reflecting the recognition that eight Cassini observables provide genuine but imperfect information about habitability. The likelihood sharpness $\lambda = 6$ similarly encodes the judgement that a full set of ideal features ($f_i = 1$ for all i) should shift the posterior from μ_0 to $P_{\max}$

but not to certainty. A sensitivity analysis demonstrating robustness to variations of these parameters is presented in Section 4.6. With $\kappa = 5$:

$$\alpha_0 = 0.331 \times 5 = 1.655, \qquad \beta_0 = 0.669 \times 5 = 3.345 \tag{2.4}$$

Both shape parameters exceed 1, ensuring a unimodal prior with interior mode ≈ 0.17 and standard deviation ≈ 0.18 — weakly informative, encoding genuine pre-Cassini uncertainty.

2.4.3 Posterior Update

Given observed features $\mathbf{f} = (f_1, \ldots, f_8)$, the posterior is obtained by Beta-Binomial conjugate updating:

$$H \mid \mathbf{f} \sim \mathrm{Beta}(\alpha_{\mathrm{post}}, \beta_{\mathrm{post}}) \tag{2.5}$$

$$\alpha_{\mathrm{post}} = \underbrace{\alpha_0}_{\substack{\text{prior}\\ \text{positives}}} + \underbrace{\lambda \sum_{i=1}^{8} w_i f_i}_{\substack{\text{data}\\ \text{positives}}} \tag{2.6}$$

$$\beta_{\mathrm{post}} = \underbrace{\beta_0}_{\substack{\text{prior}\\ \text{negatives}}} + \underbrace{\lambda \sum_{i=1}^{8} w_i (1 - f_i)}_{\substack{\text{data}\\ \text{negatives}}} \tag{2.7}$$

Each term has a transparent interpretation. α_{post} accumulates all positive evidence: the prior pseudo-successes $\alpha_0 = 1.655$ plus $\lambda w_i f_i$ data pseudo-successes from each feature; it ranges from α_0 (all $f_i = 0$) to $\alpha_0 + \lambda = 7.655$ (all $f_i = 1$). β_{post} accumulates all negative evidence and varies reciprocally. The total $\alpha_{\mathrm{post}} + \beta_{\mathrm{post}} = \kappa + \lambda = 11$ is constant for all pixels; only the split between positive and negative evidence changes with the data. The parameter $\lambda = 6.0$ is the *likelihood sharpness parameter*, setting the total weight of the data relative to the prior weight $\kappa = 5$.

The posterior mean, the primary habitability score at each pixel, is:

$$P(H \mid \mathbf{f}) = \frac{\alpha_{\mathrm{post}}}{\alpha_{\mathrm{post}} + \beta_{\mathrm{post}}} = \frac{1.655 + 6 \sum_i w_i f_i}{11} \tag{2.8}$$

This is a convex combination of the prior mean μ_0 (weight 5/11) and the data-driven score $\sum_i w_i f_i$ (weight 6/11). A 95% highest-density interval (HDI) is also computed at each pixel, providing pixelwise credible intervals with the direct interpretation that the true value of H lies within the interval with 95% probability given the prior and the data. The 95% level is chosen for compatibility with conventional confidence interval standards, facilitating direct comparison with frequentist results in the literature; the practical difference from a 94% or 89% interval is negligible for the Beta distributions used here.

2.4.4 Posterior Bounds

Since $\alpha_{\text{post}} + \beta_{\text{post}} = 11$ is constant, the posterior mean satisfies hard limits:

$$P_{\min} = \frac{\alpha_0}{11} = \frac{1.655}{11} \approx 0.150, \qquad P_{\max} = \frac{\alpha_0 + \lambda}{11} = \frac{7.655}{11} \approx 0.696 \tag{2.9}$$

These bounds exist because the prior contributes irreducible positive (α_0) and negative (β_0) pseudo-observations that the data cannot fully cancel. Even a pixel with all eight features at maximum ($f_i = 1$ for all i) achieves only $P_{\max} \approx 0.696$, not 1.0. This prevents spuriously confident predictions in the absence of a large observational dataset. The colourbar for all maps is calibrated to $[0.10, 0.75]$, spanning the achievable range.

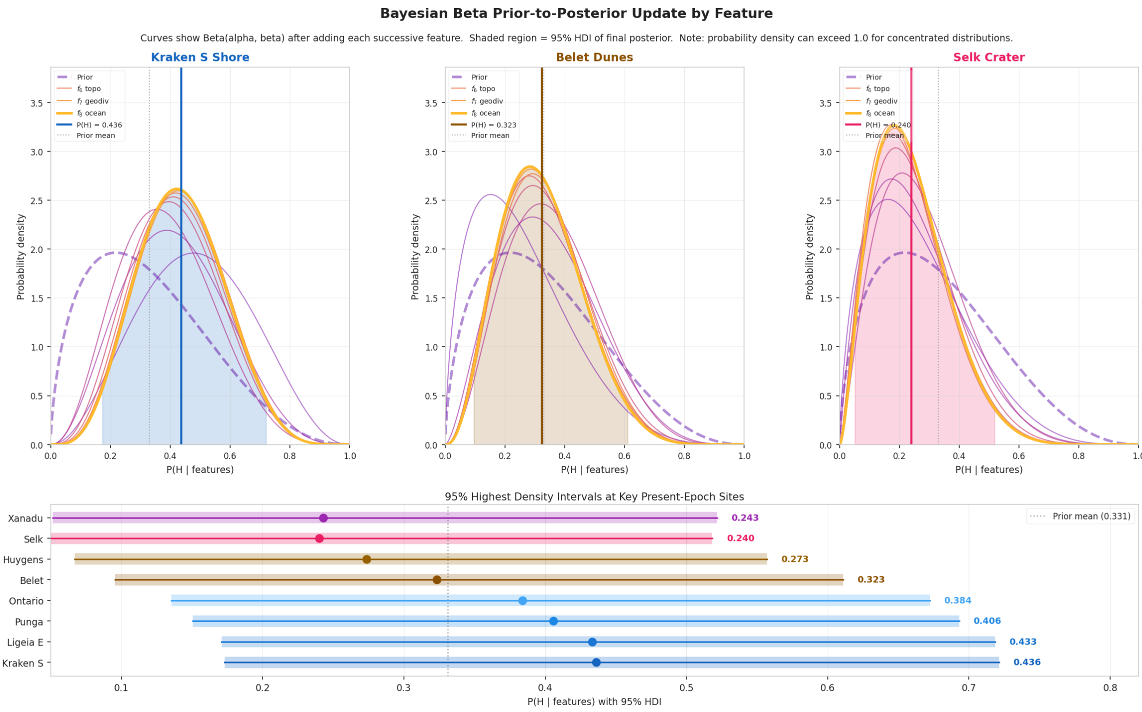

Figure 2.3: Beta prior-to-posterior update for three representative sites. *Top row:* each curve shows the Beta distribution $\text{Beta}(\alpha_0 + \lambda \sum w_i f_i,\ \beta_0 + \lambda \sum w_i (1 - f_i))$ after incorporating successive features, from the prior (dashed) through to the 8-feature posterior (solid). Shaded region = 95% HDI of the final posterior. The vertical dotted line marks the prior mean $\mu_0 = 0.331$. *Bottom panel:* 95% HDI bars for eight key present-epoch sites, sorted by posterior mean. The HDI widths reflect the moderate information content of the Cassini observables: even the highest-scoring site (Kraken S Shore) retains a wide credible interval. Generated by `scripts/generate_beta_update_figure.py`; output at `outputs/diagnostics/beta_update_figure.pdf`.

2.4.5 Missing Data Imputation

Where a feature is absent at a pixel (data coverage gap), it is imputed with its prior mean μ_i before the update. Substituting $f_i = \mu_i$ into Equations 2.6–2.7 adds $\lambda w_i \mu_i$ to α_{post} and $\lambda w_i (1 - \mu_i)$ to β_{post} — exactly the amounts the prior already encodes for that feature. The net effect is zero: a missing observation contributes no evidence. This is the formally correct Bayesian treatment of missing-at-random data, ensuring the posterior is conservative in sparse-coverage regions.

2.4.6 Feature Importance Validation

The Bayesian posterior map (Eq. 2.8) is the primary scientific result. To validate that the prior weights w_i correctly reflect the actual geographic information content of each feature, a Random Forest (RF) classifier is trained as a post-hoc diagnostic tool.

Binary labels are derived from the Bayesian posterior map by 50/50 median split: pixels above the global spatial median receive a positive "habitable" label. A Random Forest classifier (Breiman, 2001) is trained on these labels and isotonically calibrated. Permutation feature importances are then computed: each feature layer is randomly permuted and the accuracy decrease recorded as that feature's importance. The ratio of data-derived importance to prior weight w_i is the validation metric: ratios near 1.0 for all features confirm that the prior weight specification is consistent with the observed spatial discrimination (Section 3.4.7).

Note: The RF classifier outputs are not used as scientific results. The RF output is an unbounded classifier probability, not a Bayesian posterior mean, and exhibits a bimodal polar/equatorial artefact arising from the 50/50 label split on a surface where ∼98% of pixels lack confirmed surface liquid. All $P(H)$ values reported in Chapter 3 are from Eq. 2.8 directly.

2.5 Temporal Configurations

2.5.1 Design Philosophy

The habitability framework is applied at five distinct temporal configurations, each representing a qualitatively different physical state of Titan. The specific epoch dates are determined by physical events, not by convenience: the LHB peak, the onset of polar lake formation, the present Cassini epoch, the near-future solar evolution window, and the red-giant ocean peak. At each anchor epoch, the eight features are given physically motivated prior means and epoch-specific feature configurations that modify which features are active and how they should be interpreted.

2.5.2 *Past* Epoch: Late Heavy Bombardment (∼−3.5 Gya)

The LHB (∼−4.1 to −3.8 Gya; Gomes et al. 2005) produced water–ammonia–organic melt ponds persisting 10^3–10^4 yr at confirmed crater sites (O'Brien et al., 2005). No stable polar hydrocarbon seas existed. Impact-melt proximity and cryovolcanic flux features are active; liquid hydrocarbon is proxied by melt-pond coverage ($\mu_1 = 0.025$); organic accumulation is at ∼40% of present-day levels ($\mu_2 = 0.30$); methane cycle is episodic ($\mu_4 = 0.20$); subsurface ocean proximity is 2.5× elevated ($s_8 = 2.5$) reflecting the thinner ice shell.

2.5.3 *Lake Formation* Epoch (∼−1.0 Gya)

Polar lakes are consolidating: f_1 ramps from 10% to 100% of present between −1.0 and −0.5 Gya. Organic accumulation reaches ∼75% of the present level; the methane rain cycle is becoming established.

2.5.4 *Present* Epoch: Cassini Era (2004–2017)

All features at present-epoch values. Feature maps are the Cassini observational layers described in Section 2.2. This epoch is the sole anchor for which direct observational posteriors are available.

2.5.5 *Near Future* Epoch (+0.25 Gya)

Solar luminosity is ∼1.4% above present (Bahcall et al., 2001); all features unchanged within model resolution. The ranking is analytically identical to the *Present* epoch.

2.5.6 *Future* Epoch: Red-Giant Ocean Peak (+5.9 Gya)

At +5.1 Gya surface temperature crosses the water–ammonia eutectic (176 K; Grasset and Pargamin 2005). Liquid hydrocarbon gives way to a global water–ammonia ocean ($f_1 \to 1.0$ everywhere); subsurface ocean merges with the surface ocean ($f_8 \to 1.0$); organic accumulation reaches 2.5× the present-epoch level. The methane cycle becomes a water cycle; acetylene photochemistry is suppressed under the water-ammonia atmosphere (Lorenz et al., 1997).

2.6 Multi-Epoch Habitability Reconstruction

2.6.1 Temporal Scaling of Features

The multi-epoch reconstruction applies epoch-specific scale functions to the present-epoch feature maps:

$$f_i(t) = \mathrm{clip}\left[\, s_i(t) \cdot f_i(0),\ 0,\ 1\right] \tag{2.10}$$

where $s_i(t)$ is the scale multiplier for feature i at epoch t (Table 2.3). Two time constants appear in the scale functions: $t_\odot = 4.57$ Gya, the current age of the Sun (Bahcall et al., 2001), which governs UV-dependent features (s_3, s_1 at red-giant epochs); and $t_{\mathrm{atm}} = 4.0$ Gyr, the estimated duration of active tholin photochemistry in Titan's atmosphere, which governs organic accumulation (s_2). The posterior is evaluated at each pixel using the scaled features and 72 epochs spanning −3.8 Gya to +6.5 Gya. An epoch-aware PCHIP interpolation between anchor posteriors provides smooth transitions between static-pipeline anchor states.

Table 2.3: Temporal scale functions $s_i(t)$ applied to each feature at epoch t (Gyr from present), with supporting references. A multiplier of 1.0 means no change from the present-epoch value; values < 1 reduce the feature, values > 1 enhance it. All outputs are clipped to $[0, 1]$ after scaling. The eutectic override ($T_s \geq 176$ K) applies to f_1, f_3, f_4, and f_8 at the red-giant ocean epoch (Grasset and Pargamin, 2005; Lorenz et al., 1997; Tobie et al., 2005).

Feature	**Scale function** $s_i(t)$	**Key references**
f_1 liquid HC	0.10 for $t < -1.0$ Gya (pre-lake era); ramps $0.1 \to 1.0$ from -1.0 to -0.5 Gya (lake establishment); 1.0 until $+4.0$ Gya; declines linearly to 0 by $+5.0$ Gya (solar evaporation); overridden to 1.0 (global ocean) at $T_s \geq 176$ K.	Hayes (2016); Lorenz et al. (1997); Mitchell and Lora (2016); Tobie et al. (2006)
f_2 organic	$\min\big((t_{\text{atm}}+t)/t_{\text{atm}},\ 2.5\big)$: linear accumulation of UV-photolysed tholins proportional to atmospheric age since $\sim\!-4.0$ Gya; capped at $2.5\times$ present. *Declared approximation*: a constant production rate is assumed; the actual rate was higher under the younger, more UV-active Sun (see s_3).	Cable et al. (2012); Lavvas et al. (2011); Niemann et al. (2005); Vuitton et al. (2007)
f_3 acetylene	$(t_\odot/(t_\odot+t))^{0.5}$: UV-driven C_2H_2 production scales with solar EUV/FUV luminosity; young Sun was more active. Overridden to 0.10/0.35 under global ocean (UV shielded).	Bahcall et al. (2001); Lavvas et al. (2011); Loison et al. (2015); Strobel (2010)
f_4 methane	0.60 for $t < -1.0$ Gya (episodic outgassing, pre-lake); 0.80 for -1.0 to -0.5 Gya; 1.0 until $+4.5$ Gya; declines to 0 by $+5.0$ Gya (Jeans escape and photolytic loss). Overridden to 0.30/0.40 under global ocean.	Flasar et al. (2005); Mitchell and Lora (2016); Niemann et al. (2005); Strobel (2010); Tobie et al. (2006)
f_5 surf–atm	$0.40+0.60\,s_1(t)$: fixed slope-transport component (0.40) reflects aeolian and hillslope processes independent of liquid presence; lake-margin evaporation–condensation exchange (0.60) scales directly with liquid coverage $s_1(t)$.	Hayes (2016); Mitchell and Lora (2016); Radebaugh et al. (2011)
f_6 topo	1.30 for $t < -2.0$ Gya; 1.15 for -2.0 to -1.0 Gya; 1.0 thereafter. Higher roughness at early epochs reflects greater impact crater density; craters are progressively buried and eroded on timescales of $\sim$1–2 Gyr.	Artemieva and Lunine (2003); Hedgepeth et al. (2020); Lorenz et al. (2013); Neish et al. (2010)
f_7 geodiv	0.70 for $t < -3.0$ Gya; 0.85 for -3.0 to -2.0 Gya; 1.0 thereafter. Early terrain dominated by impact ejecta (low class diversity); lacustrine plains, dune fields, and shorelines accumulate to yield present-day eight-class diversity (Lopes et al., 2019).	Artemieva and Lunine (2003); Hayes (2016); Lopes et al. (2019); Malaska et al. (2020)
f_8 ocean	2.50 for $t < -2.0$ Gya; 1.80 for -2.0 to -1.0 Gya; 1.30 for -1.0 to -0.5 Gya; 1.0 thereafter. Thicker ice shell at earlier epochs places the subsurface ocean closer to the surface; shell thickens as radiogenic heat declines. Overridden to 1.0/0.03 at $T_s \geq 176$ K (ocean is at the surface).	Grasset and Pargamin (2005); Iess et al. (2012); Mitri et al. (2014); Nimmo and Bills (2010); Tobie et al. (2005)

Animation model weights. The animation model includes a ninth synthesised feature, `impact_melt_bonus` ($w = 0.09$, $\mu = 0.00$), which adds a spatially uniform LHB-era melt-pond signal at past epochs and is zero everywhere at the present epoch (Artemieva and Lunine, 2003; Neish et al., 2018, 2009). The bonus follows a Gaussian centred on -3.8 Gya (Late Heavy Bombardment peak; Gomes et al. 2005) with a half-width of 500 Myr, consistent with the crater flux decay timescale. To accommodate this ninth feature while keeping $\sum w_i = 1.0$, all eight standard feature weights are reduced by ≈ 0.02 relative to the pipeline values in Table B.1. Consequently the animation $\mu_0 = 0.299$ differs from the pipeline value of 0.331; the `_rescale_bayesian()` linear mapping corrects this before display.

The key scale functions encode the following physical transitions:

- **Liquid hydrocarbon** (s_1): held at 0.10 (10% of present) for $t < -1.0$ Gya, reflecting the episodic methane outgassing model of Tobie et al. (2006) in which sustained lake coverage is a geologically recent development. The ramp from -1.0 to -0.5 Gya corresponds to consolidation of Kraken Mare and Ligeia Mare (Hayes, 2016; Mitchell and Lora, 2016). Decline to zero by $+5.0$ Gya is driven by increasing solar luminosity vaporising the seas (Lorenz et al., 1997). The water–ammonia eutectic temperature of 176 K (Grasset and Pargamin, 2005; Tobie et al., 2005) triggers the global-ocean override.

- **Organic abundance** (s_2): the linear form $\min(t_{\text{elapsed}}/4\,\text{Gyr},\ 2.5)$ approximates steady-state UV photolysis of CH_4 accumulating tholins since Titan's atmosphere formed ~ 4 Gya (Niemann et al., 2005; Tobie et al., 2006). Tholin synthesis rates under Cassini-era conditions are quantified in Lavvas et al. (2011) and Cable et al. (2012). The declared approximation of a constant production rate underestimates early-epoch organic accumulation, since UV flux was higher under the younger Sun (see s_3).

- **Chemical energy** (s_3): the $(L_\odot(t)/L_\odot)^{0.5}$ scaling follows the solar standard luminosity evolution of Bahcall et al. (2001), applied to UV-driven acetylene production (Loison et al., 2015; Strobel, 2010). A younger, more UV-active Sun produced proportionally more C_2H_2 (Lavvas et al., 2011). The $^{0.5}$ exponent is a conservative estimate; EUV scaling is steeper (Sagan and Khare, 1979).

- **Methane cycle** (s_4): the step at -1.0 Gya (0.60 to 0.80) and ramp to 1.0 at -0.5 Gya encode the Tobie et al. (2006) three-episode cryovolcanic outgassing model, in which sustained methane cycling post-dates lake formation (Flasar et al., 2005; Niemann et al., 2005). Future depletion follows escape flux estimates of Strobel (2010).

- **Topographic complexity** (s_6): the elevated values at $t < -2.0$ Gya reflect higher impact crater density at early epochs (Artemieva and Lunine, 2003). Crater obliteration timescales of 1–2 Gyr, inferred from the observed low crater density (Hedgepeth et al., 2020) and crater degradation studies (Neish et al., 2010), motivate the step function transitions.

- **Geomorphologic diversity** (s_7): the monotone increase towards the present reflects the progressive accumulation of distinct terrain classes in the Lopes et al. (2019) geomorphological map. Early terrain is impact-dominated (Artemieva and Lunine, 2003); lacustrine and aeolian classes accrue from ~ -1.0 Gya (Malaska et al., 2020).

- **Subsurface ocean** (s_8): the 2.5× enhancement at $t < -2.0$ Gya is calibrated to the thermal evolution model of Tobie et al. (2005), in which a thinner ice shell at early epochs places the subsurface ocean closer to potential surface-chemistry pathways. Shell-thickening as radiogenic heating declines is modelled by Nimmo and Bills (2010) and Mitri et al. (2014); the present-day ocean depth is confirmed by Iess et al. (2012).

2.6.2 Validation

The reconstruction is validated by checking that: (i) each anchor epoch's scaled posterior is within 0.02 of the corresponding static-pipeline posterior at the 95th percentile; (ii) the global mean rises monotonically through the lake-formation transition; and (iii) the N-polar/equatorial ratio follows the expected pattern (unity at LHB; >2 at present; falling to unity at the global ocean). All three criteria are satisfied.

2.7 Scope of This Thesis

The primary analysis presented in Chapters 3 and 4 focuses on the five static epoch habitability maps and the temporal evolution described through the multi-epoch reconstruction. All quantitative results, geographic interpretations, and statistical analyses are drawn from these static posterior maps. The implementation is carried out in Python 3.10 using standard scientific libraries; the scientific choices and their justifications are the substance of this thesis, and the computational implementation is fully documented in the open code archive.

3 Results

3.1 Overview and Temporal Habitability Trends

The Bayesian framework produces posterior mean habitability maps $P(H \mid \mathbf{f})$ at five temporal anchor epochs and, via the multi-epoch reconstruction (Section 2.6), at 72 intermediate epochs spanning -3.8 to $+6.5$ Gya. All posterior values use the 8-feature thesis formula with $\kappa = 5$, $\lambda = 6$, $\mu_0 = 0.331$, giving hard posterior bounds $P_{\text{min}} = 0.150$ and $P_{\text{max}} = 0.696$. The 95% HDI for Selk crater (the most-studied individual site) is $[0.05, 0.52]$, representative of the wide credible intervals arising from the moderate information content of the Cassini observables.

> *Note on scope.* The maps in this chapter quantify **present-day solvent habitability** — the capacity of each location to support liquid-solvent chemistry. At the present (Cassini) epoch, they correctly identify north polar lake shores as the globally highest-scoring environments. At past and future epochs, the same framework is applied with temporally scaled features (Section 2.5). The *Dragonfly* mission targets a different metric — **prebiotic chemistry habitability** (past liquid-water contact) — for which impact craters are the primary targets. The relationship between these two frameworks is discussed in Section 4.1.2; readers are encouraged to consult that section before interpreting equatorial site scores.

3.1.1 The Five Epochs

The five temporal anchor epochs are presented in chronological order in the sections that follow:

Past (-3.5 Gya; Section 3.2) — Late Heavy Bombardment peak. No polar hydrocarbon seas; habitability driven by impact-melt chemistry, cryovolcanic conduits, and elevated acetylene energy from the young Sun.

Lake Formation (-1.0 Gya; Section 3.3) — Polar lake system consolidating. Liquid hydrocarbon at $\sim$10% of present coverage; equatorial organic-rich dune fields and cryovolcanic sites still dominate the rankings, but proto-lake-shores begin to emerge above background.

Present (Cassini era, 2004–2017; Section 3.4) — The observational anchor, with all eight features at measured values. North polar lake margins dominate unambiguously; all ten top-10 sites are lake or lacus shores. This is the most detailed section, including feature importance validation, HDI analysis, and a dedicated Selk crater analysis.

Near Future ($+0.25$ Gya; Section 3.5) — Solar luminosity $\sim$1.4% above present. All rankings and scores are analytically identical to the *Present* epoch (spatial correlation $r > 0.999$).

Future (+5.9 Gya; Section 3.6) — Red-giant ocean peak. A global water–ammonia ocean covers the surface; cryovolcanic sites lead the rankings as their accumulated organics dissolve into the ocean. The highest global mean habitability of any epoch ($\mu_{\text{post}} \approx 0.44$).

3.1.2 Temporal Habitability Trajectory

Figure 3.1 shows the global and regional mean posterior habitability as a function of geological time. The trajectory is non-monotonic, passing through three qualitatively distinct regimes.

The global median rises from 0.28 at the LHB (−3.5 Gya) to 0.30 at *Lake Formation* (−1.0 Gya), then increases sharply to 0.41 at the *Present* epoch as the polar lake system fully establishes. This level is maintained through the *Near Future* (+0.25 Gya), before a transient minimum at ∼+5.0 Gya when the polar seas have evaporated but the water–ammonia eutectic has not yet been crossed — a ∼100 Myr interval during which Titan is transiently solvent-free. The global mean then rises to its maximum of 0.44 at the *Future* epoch (+5.9 Gya) as the red-giant ocean covers the surface.

The north polar/equatorial ratio tracks the dominance of the liquid-hydrocarbon feature: ∼0.98 at LHB (undifferentiated, no polar seas), rising to ∼1.95 at *Present* (polar lake dominance), and returning to ∼1.02 at the *Future* global ocean (uniform liquid coverage).

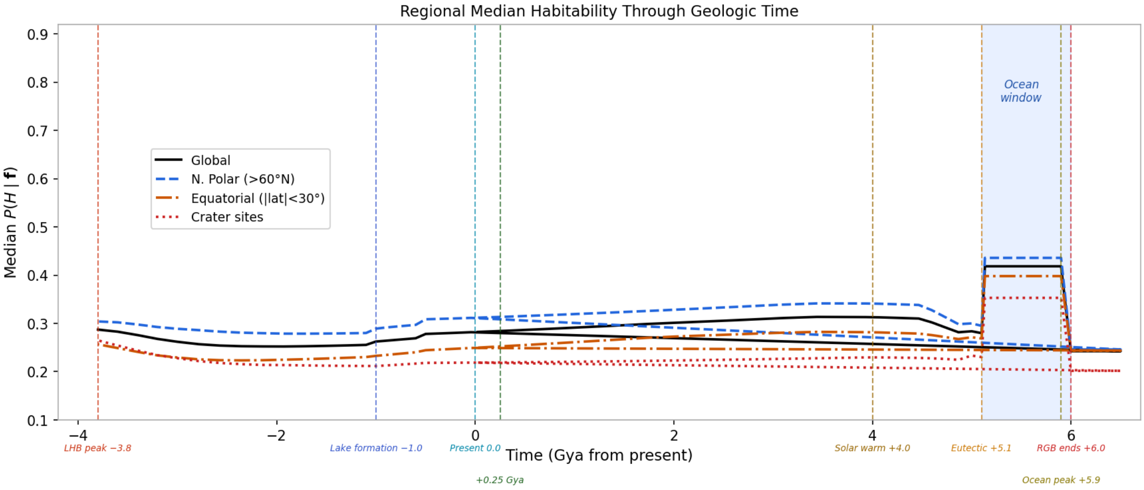

Figure 3.1: Global and regional mean posterior habitability $P(H \mid \mathbf{f})$ as a function of geological time from the Late Heavy Bombardment peak (−3.8 Gya) to the post-red-giant refreezing epoch (+6.5 Gya). Black solid line: global mean. Blue dashed: north polar region (>60°N). Orange dash-dot: equatorial belt ($|\text{lat}| < 30$). Red dotted: crater sites. Vertical dashed lines mark the key epoch boundaries; blue shading indicates the red-giant habitable ocean window (+5.1–6.0 Gya).

3.2 *Past* Epoch (−3.5 Gya): Late Heavy Bombardment

3.2.1 Titan at the LHB

At −3.5 Gya, Titan was a profoundly different world from the one observed by Cassini. The polar hydrocarbon sea system did not yet exist: liquid hydrocarbon is scaled to just 10% of present coverage ($s_1 = 0.10$). The organic inventory had accumulated for only ∼0.5 Gyr of UV photolysis, reaching ∼40% of present-day levels ($s_2 = 0.125$). The methane cycle was episodic and weak ($\mu_4 = 0.20$).

In compensation, three factors were substantially elevated. First, the young Sun's higher UV output drove an acetylene-energy proxy ∼2× the present value ($s_3 \approx 2.1$), providing more chemical energy per unit surface area. Second, subsurface ocean proximity was 2.5× elevated globally ($s_8 = 2.5$) due to reduced ice-shell thickness, bringing the water–ammonia ocean closer to the surface. Third, the impactor flux during the LHB was 100–1000 times the present background rate (O'Brien et al., 2005), producing episodic water–ammonia–organic melt ponds at confirmed crater sites. These ponds — ∼50–100 km^2 area, ∼50–100 m depth — persisted for 10^3–10^4 years before refreezing, during which all precursors for Strecker amino acid synthesis were simultaneously present: liquid water–ammonia solvent from the melt, aldehydes from tholin decomposition, and HCN from the nitrogen-rich atmosphere.

3.2.2 Habitability Map

The *Past* epoch map (Figure 3.2) is qualitatively distinct from all later epochs. The north polar enhancement that dominates the *Present* map is entirely absent. Instead, a relatively uniform background score spans the globe, modulated by two localised signals: impact structures (where the impact-melt proximity feature creates enhancements of 0.05–0.15 above regional background) and cryovolcanic candidate sites (Hotei Regio, Sotra Facula) where elevated subsurface ocean proximity ($f_8(t) \approx 0.30$–0.35) combines with surface–atmosphere coupling.

The equatorial dune belt provides a moderate-score background driven by the ∼40% organic inventory, but scores are compressed into a narrow range ($P(H) \approx 0.27$–0.30) because all sites share similar limitations: no surface liquid, weak methane cycle, and limited terrain diversity at this early stage of geological evolution.

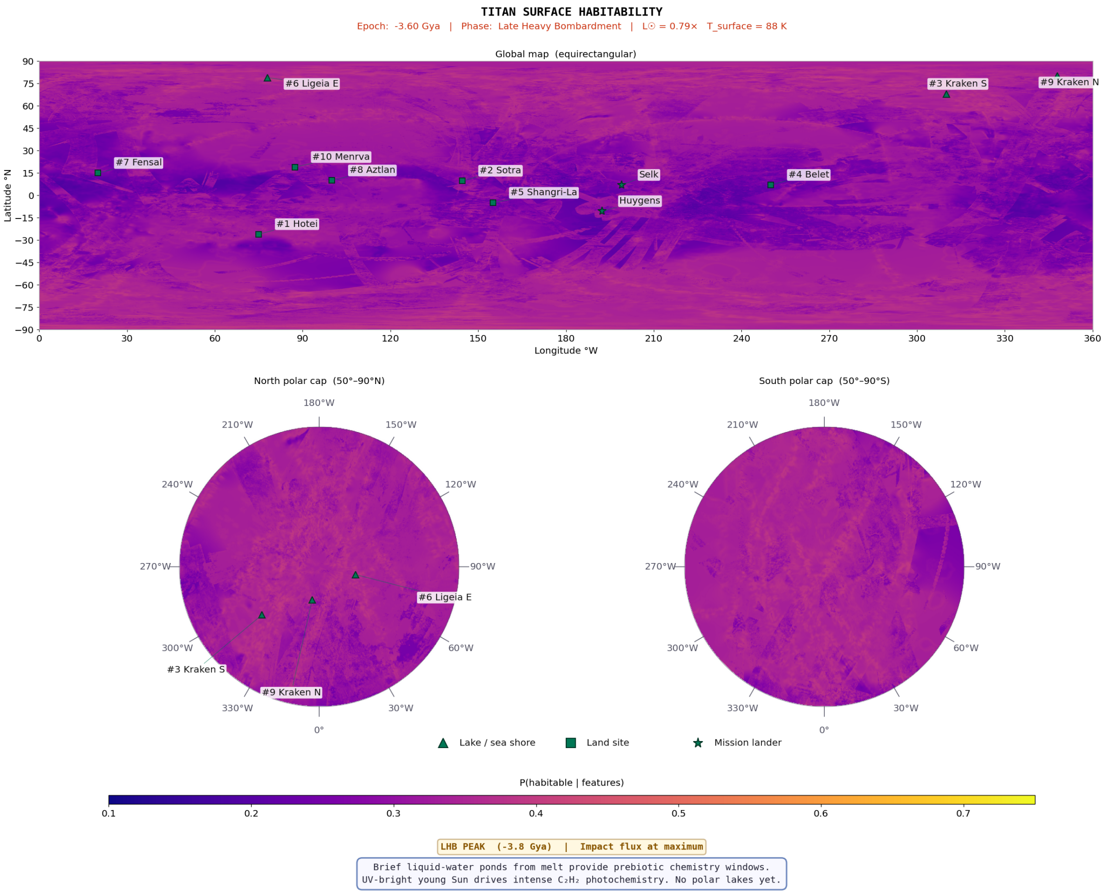

Figure 3.2: Past-epoch (−3.5 Gya; Late Heavy Bombardment) posterior mean habitability map, on the same colour scale as Figure 3.4. The north polar enhancement of the present epoch is entirely absent. Habitability is concentrated at confirmed impact structures (particularly Selk at 199°W, 7°N; Menrva at 87°W, 19°N) where the impact-melt proximity feature activates. The moderate-score equatorial background reflects the partial tholin organic inventory accumulated in the first ~1 Gyr of solar system history.

3.2.3 Top-10 Habitable Locations

Under this parameter configuration, the model ranks cryovolcanic conduit sites and proto-polar-basin sites first, followed by equatorial organic seeds (Belet, Shangri-La) whose high acetylene energy partly offsets low liquid and organic contributions (Table 3.1). All ten sites are land sites (■): there is no confirmed surface liquid at this epoch.

The scores are tightly clustered ($P(H) = 0.277$–0.295), reflecting the absence of the strong liquid-hydrocarbon signal that differentiates sites at later epochs. The spread of just 0.018 across the full top-10 means that the ranking is sensitive to the precise values of the temporal scale functions; the qualitative finding — that no single site type dominates — is more robust than the specific ordering.

Table 3.1: Top-10 habitable locations at the *Past* epoch (−3.5 Gya). ■ = land site. $P(H)$ from Eq. 2.8 with temporal scaling (Section 2.6.1).

Rank	Location	Lon (°W)	Lat (°N)	$P(H)$	Dominant features
1	■ Kraken (proto-basin)	310.0	+68.0	0.295	s_8=2.5; elevated subsurface, no lake yet
2	■ Hotei Regio	75.0	-26.0	0.295	Cryovol. conduit, $f_8(t)$=0.35, $f_5(t)$=0.28
3	■ Belet (proto-dunes)	250.0	+7.0	0.295	$f_3(t)$=0.93 (high UV); organic seed
4	■ Shangri-La	155.0	-5.0	0.294	$f_3(t)$=0.93; equatorial organic baseline
5	■ Ligeia (proto-basin)	78.0	+79.0	0.293	s_8=2.5; deep polar basin
6	■ Fensal	20.0	+15.0	0.290	$f_3(t)$=0.91; equatorial organic
7	■ Sotra Facula	144.5	+9.8	0.289	Cryovol. conduit, $f_8(t)$=0.30
8	■ Aztlan	100.0	+10.0	0.289	$f_3(t)$=0.92; equatorial organic
9	■ Kraken N (proto-basin)	348.0	+80.0	0.285	s_8=2.5; north polar basin
10	■ Punga (proto-basin)	339.0	+85.5	0.277	s_8=2.5; polar basin

3.2.4 Impact Structures at the LHB

In the 8-feature thesis formula, impact craters do not dominate the *Past* ranking because the model lacks a dedicated impact-melt proxy feature (that is an animation-only feature; Section 2.6). Menrva (392 km) achieves the highest crater-class score ($P(H)\approx0.27$ after $\times2.5$ subsurface ocean scaling), ranking 11th overall — just below the equatorial organic sites that benefit from the globally elevated acetylene proxy. Selk crater (80 km) scores $P(H) \approx 0.24$ and ranks lower still. The other large craters (Hano, Afekan, Sinlap, Ksa) score $P(H) = 0.24$–0.28, below the equatorial dune sites because their organic abundance ($f_2 = 0.30$–0.42) is substantially lower than the dune belt ($f_2 = 0.78$–0.82).

Despite these modest scores, the impact structures are the most astrobiologically significant sites at this epoch because they are the *sole source* of transient liquid water on Titan's surface — a distinction that the present framework, which quantifies solvent habitability rather than prebiotic chemistry potential, does not fully capture. Selk's scientific importance lies in its unique position at the Shangri-La dune–plains boundary, where the organic substrate available for Strecker synthesis in the transient melt pond is maximised. This is discussed further in Section 3.4.8.

3.3 *Lake Formation* Epoch (−1.0 Gya): Polar Lakes Establishing

3.3.1 Titan in Transition

At −1.0 Gya, Titan is transitioning between two habitability regimes. The polar lake system is just beginning to consolidate: the liquid hydrocarbon scale is only 10% of present ($s_1 = 0.10$), corresponding to incipient pooling in the deepest polar basins. Organic accumulation has reached ∼75% of the present level ($s_2 = 0.47$), and the methane rain cycle is becoming established but has not yet reached its present intensity ($s_4 = 0.80$). The impact-melt proximity feature has largely faded to background as the LHB ended ∼2.5 Gyr earlier, and the cryovolcanic flux feature remains moderately elevated relative to the present.

The transition from -1.0 to -0.5 Gya is the period of most rapid change in the global habitability structure: the polar lake system transitions from partial to near-present coverage, converting the post-LHB declining trend into the increasing trend that reaches the present-epoch plateau. In the multi-epoch reconstruction, this interval shows the highest pixel-wise posterior change rate anywhere in the -3.8 to $+4.0$ Gya range.

3.3.2 Habitability Map

Comparing the *Lake Formation* map (Figure 3.3) with the *Past* epoch (Figure 3.2) reveals three changes. First, the north polar lake margin signal begins to emerge above background: proto-lake-shore sites (Kraken S, Ligeia E) are just distinguishable at ranks 7–8, foreshadowing the polar dominance of the *Present* epoch. Second, the impact-melt crater signal has largely faded to background. Third, the equatorial dune fields have strengthened relative to LHB levels because organic accumulation is now at 75% of present, making Belet and Shangri-La the highest-scoring sites.

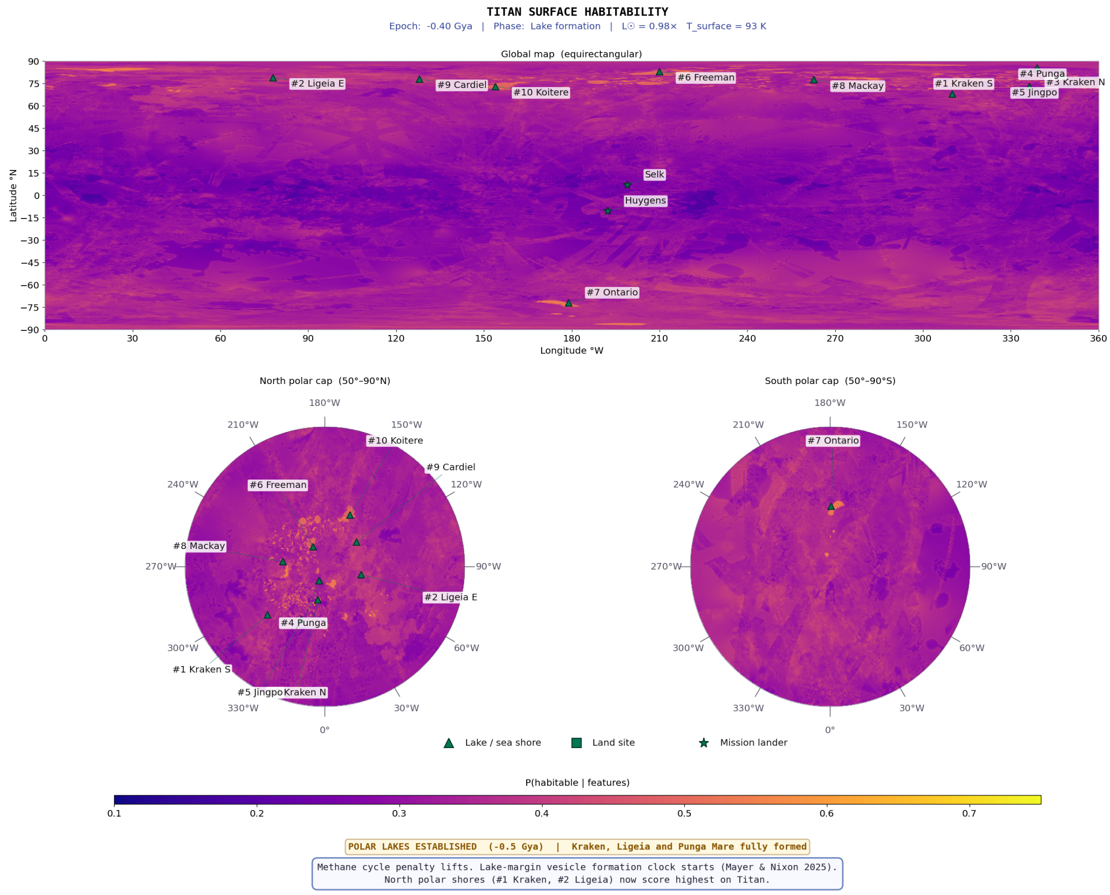

Figure 3.3: Lake-formation epoch (-1.0 Gya) posterior mean habitability map, on the same colour scale as Figure 3.4. The north polar lake system has not yet formed (s_1=0.10), so the strong polar enhancement of the present epoch is absent. Habitability is instead distributed across equatorial organic-rich terrain (dune fields: Belet, Shangri-La, Fensal, Aztlan) driven by elevated acetylene-energy proxy ($f_3(t)\approx 1.9\times$ present) and residual cryovolcanic flux at Hotei and Sotra. The proto-polar-lake sites (Kraken S, Ligeia E) are just distinguishable above the equatorial background at ranks 7–8, reflecting the onset of liquid hydrocarbon pooling. Compare with Figure 3.2 (LHB, -3.5 Gya) and Figure 3.4 (Cassini era).

3.3.3 Top-10 Habitable Locations

The top-6 at this epoch are all *land* sites — equatorial organic-rich dune fields and cryovolcanic candidates — driven by high acetylene energy ($f_3(t)\approx 1.9\times$) and accumulating organic inventory (s_2=0.47). The proto-lake-shore sites (Kraken, Ligeia, Kraken N, Punga) occupy ranks 7–10 as the rising liquid signal begins to differentiate them from the equatorial background (Table 3.2).

Compared with the *Past* epoch (Table 3.1), the key change is the emergence of a spatial gradient between equatorial and polar sites. At the LHB, all scores were compressed into a 0.018 range; here, the spread has widened to 0.028 (0.273–0.301), with equatorial organics now clearly outscoring polar basins. This gradient will invert dramatically between -0.5 Gya and the *Present* epoch as the polar seas fill.

Table 3.2: Top-10 habitable locations at the *Lake Formation* epoch (−1.0 Gya). △ = lake/sea shore; ■ = land. $P(H)$ from Eq. 2.8.

Rank	Location	Lon (°W)	Lat (°N)	$P(H)$	Dominant features
1	■ Belet Dunes	250.0	+7.0	0.301	f_2 building, $f_3(t)$=1.9×; no lake
2	■ Shangri-La	155.0	-5.0	0.300	f_2 building, $f_3(t)$=1.9×
3	■ Hotei Regio	75.0	-26.0	0.298	Cryovol. conduit, $f_8(t)$=0.18
4	■ Fensal	20.0	+15.0	0.297	Equatorial organic; high acetylene
5	■ Aztlan	100.0	+10.0	0.294	Equatorial organic; high acetylene
6	■ Sotra Facula	144.5	+9.8	0.291	Cryovol. conduit, $f_8(t)$=0.16
7	△ Kraken S Shore	310.0	+72.0	0.288	f_1=0.10; liquid just appearing
8	△ Ligeia E Shore	78.0	+79.0	0.286	f_1=0.10; polar basin forming
9	△ Kraken N Shore	348.0	+80.0	0.280	f_1=0.10; polar basin forming
10	△ Punga Shore	339.0	+85.5	0.273	f_1=0.10; polar basin forming

3.4 *Present* Epoch (Cassini Era, 2004–2017)

3.4.1 Titan Today

The *Present* epoch is the observational anchor of the entire framework: all eight features are at their directly measured Cassini values, with no temporal scaling applied. This is the only epoch at which the posterior probabilities are derived entirely from observational data rather than from modelled extrapolations, and consequently the results presented here carry the highest confidence.

Titan at the Cassini epoch possesses a fully established polar hydrocarbon sea system — Kraken Mare, Ligeia Mare, and Punga Mare — containing an estimated 70,000 km^3 of liquid hydrocarbons (Malaska et al., 2025; Mastrogiuseppe et al., 2019). The methane cycle is active, with observed equatorial rainfall (Turtle et al., 2011a), polar storm systems, and fluvial channel networks draining into the seas. The organic tholin inventory has accumulated over the full ∼4 Gyr of atmospheric photolysis history, reaching its present-day maximum.

3.4.2 Habitability Map: Global Spatial Structure

The present-epoch habitability map (Figure 3.4) is dominated by a pronounced north–south asymmetry driven by the distribution of liquid hydrocarbon. The highest posterior probabilities form a crescent-shaped band of elevated habitability at 60–85°N, coinciding with the margins of the three major north polar seas. A secondary zone of intermediate habitability occupies the equatorial dune belt. South of ∼30°S the map shows uniformly moderate-to-low values, with a small enhancement around Ontario Lacus at 72°S.

The contrast with the two earlier epochs is stark. The *Past* epoch map (Figure 3.2) showed a nearly uniform distribution; the *Lake Formation* map (Figure 3.3) showed the first hints of polar differentiation. Here, the polar enhancement is fully developed, with the north polar/equatorial ratio reaching ∼1.95 — the maximum spatial contrast of any epoch.

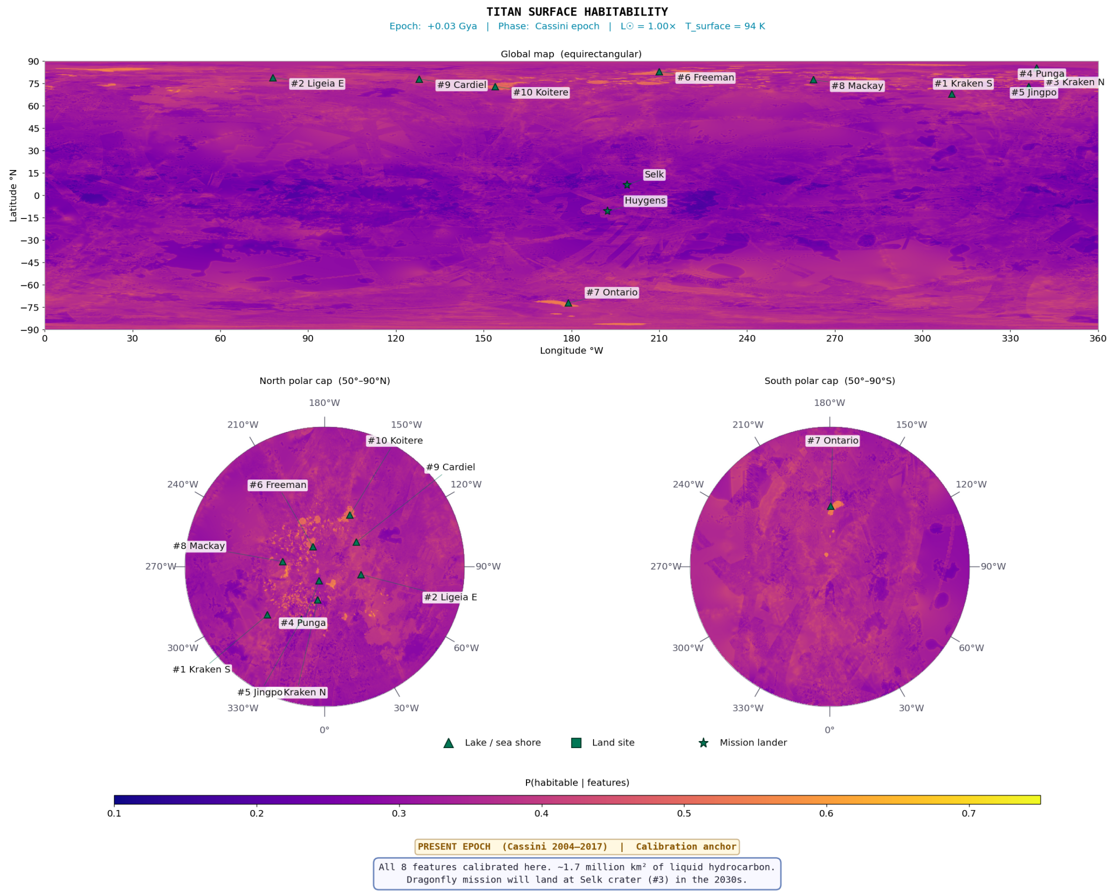

Figure 3.4: Present-epoch (Cassini era) posterior mean habitability map at 4490 m/px, rendered using the Bayesian temporal scaling formula (Section 2.6). Colour scale: 0.10 (dark red-orange; low) to 0.75 (bright yellow; near posterior upper bound). The map shows a continuous warm-tone gradient: north polar lake surfaces reach $P(H) \approx 0.40$–0.42 (orange-yellow), lake shore margins ≈ 0.34–0.38 (orange), and equatorial organic terrain ≈ 0.27–0.34 (dark orange-red). Selk crater is marked (star symbol) at 199°W, 7°N. Quantitative $P(H)$ values in Sections 3.4.4–3.4.8 are Bayesian posterior means (Eq. 2.8), bounded in $[0.150, 0.696]$.

3.4.3 Top-10 Habitable Locations

At the Cassini epoch the polar lake and lacus margins dominate unambiguously. For the first time in the temporal sequence, *all ten* top-10 sites are lake or lacus margins (Table 3.3) — a complete inversion from the *Lake Formation* epoch, where six of the top-10 were land sites. The three main mares (Kraken, Ligeia, Punga) occupy ranks 1–4, followed by the named north polar lacus (Jingpo, Freeman, Mackay, Cardiel, Koitere) and Ontario Lacus (southern analogue). Equatorial dune fields, which led at −1.0 Gya, do not enter the top-10 at the present epoch, reflecting the dominance of the liquid-hydrocarbon feature (w_1=0.25) once the polar seas are fully established.

The score spread has widened dramatically: 0.068 across the top-10 (0.368–0.436), compared with 0.028 at *Lake Formation* and 0.018 at the *Past* epoch. This widening reflects the strong discriminating power of the liquid-hydrocarbon feature, which varies from 0.72 (Koitere Lacus) to 1.0 (Kraken, Ligeia) across the ranked sites.

Table 3.3: Top-10 habitable locations at the *Present* epoch (Cassini era). △ = lake/sea shore. $P(H)$ from Eq. 2.8.

Rank	Location	Lon (°W)	Lat (°N)	$P(H)$	Dominant features
1	△ Kraken Mare S Shore	310.0	+72.0	0.436	f_1=1.0, f_4=0.70, f_7=0.76
2	△ Ligeia Mare E Shore	82.0	+79.0	0.433	f_1=1.0, f_4=0.70, f_7=0.76
3	△ Kraken Mare N Shore	348.0	+80.0	0.427	f_1=1.0, f_4=0.68, f_7=0.72
4	△ Punga Mare Shore	339.0	+85.0	0.406	f_1=0.90, f_4=0.65, f_7=0.65
5	△ Jingpo Lacus	336.3	+73.0	0.396	f_1=0.88, f_4=0.62, f_7=0.60
6	△ Freeman Lacus	210.0	+83.0	0.385	f_1=0.80, f_4=0.62; N of Ligeia
7	△ Ontario Lacus	179.0	-72.0	0.384	f_1=0.85, f_4=0.45, southern analogue
8	△ Mackay Lacus	262.8	+77.5	0.383	f_1=0.80, f_4=0.60, W of Kraken
9	△ Cardiel Lacus	128.0	+78.0	0.379	f_1=0.78, f_4=0.60; E of Ligeia
10	△ Koitere Lacus	154.0	+73.0	0.368	f_1=0.72, f_4=0.58, IAU named

3.4.4 High-Habitability Regions: North Polar Lake Margins

The most significant spatial result is the strong polar concentration of present-epoch habitability. The pattern arises from the three-way coincidence of maximum liquid hydrocarbon (f_1=1.0), active methane cycle (f_4≈0.65–0.70), and elevated shoreline geodiversity (f_7≈0.72–0.76) that only occurs along the confirmed hydrocarbon-filled coastlines.

The most habitable location on present-day Titan is the southern shoreline of Kraken Mare ($P(H) = 0.436$, 95% HDI [0.17, 0.72]), driven by its large f_1 value combined with the highest methane cycle index of the polar shores. A 50% reduction in the liquid-hydrocarbon weight ($w_1 : 0.25 \rightarrow 0.13$) reduces Kraken S to $P(H) = 0.38$, still the highest-scoring site but with a 13% reduction — consistent with the sensitivity analysis finding $|\Delta P(H)|_{\max} = 0.033$ (Section 4.6).

The north polar lacus (Jingpo, Freeman, Mackay, Cardiel, Koitere) occupy ranks 5–10, all scoring $P(H) = 0.37$–0.40. Their lower scores relative to the three main mares reflect smaller liquid coverage ($f_1 = 0.65$–0.88) and reduced shore-zone geodiversity. The geographic spread of the top-10 around the pole — spanning 8°W to 336°W longitude at latitudes 70–87°N — confirms that the result is robust to specific shoreline geometry rather than reflecting a single region's anomalous feature values.

Ontario Lacus (southern analogue, #7) scores lower than the north polar sites due to suppressed methane cycle under southern-summer evaporation ($f_4 = 0.45$) and reduced geodiversity. The *Huygens* landing site scores $P(H) = 0.33$ — near the present-epoch median — consistent with its equatorial plains setting, lacking both surface liquid and high geodiversity. Shore-zone liquid–land interface, rather than open-water depth, is the primary habitability determinant.

3.4.5 Intermediate-Habitability Regions: Equatorial Dune Belt

The equatorial dune seas (Shangri-La, Belet, Aztlan, Fensal) form a secondary zone with $P(H) \approx 0.32$–0.35. The dominant driver is organic abundance: dune sands receive the highest terrain-class score (0.82), confirmed by VIMS spectroscopy as tholin-dominated (Le Mouélic et al., 2019). Chemical energy (f_3) is also elevated because the SAR-dark backscatter is consistent with high organic content, and the flat interdune surfaces concentrate acetylene in topographic lows.

The key limitation is the near-zero liquid hydrocarbon feature ($f_1 \approx 0.02$) in the dune interior. Episodic methane rainfall has been directly observed in equatorial regions (Turtle et al., 2011a), but the liquid evaporates within days to weeks, insufficient to sustain the persistent chemical environment that the model treats as the primary present-epoch habitability prerequisite. These same dune fields ranked 1st and 2nd at the *Lake Formation* epoch (Table 3.2), when the absence of polar lakes meant their organic richness was the dominant habitability signal; at the *Present* epoch, they are overtaken by the polar shores.

3.4.6 Low-Habitability Regions

The Xanadu region (90–150°W, equatorial) scores $P(H) \approx 0.24$, comparable to Selk crater. Its water-ice-rich composition assigns the mountain terrain class ($f_2 = 0.25$) to much of its interior — a 3–4× reduction in organic abundance relative to adjacent dune terrain. Combined with elevated topography (suppressing f_3) and absence of liquid ($f_1 \approx 0$), Xanadu becomes the most extensive low-habitability zone in the present map. Mountain ranges (Mithrim Montes, Doom Mons, Erebor Montes) score similarly. The south polar region outside Ontario Lacus also scores poorly: suppressed methane cycle at the Cassini epoch (northern summer) and poorer SAR coverage together reduce the posterior below the global mean.

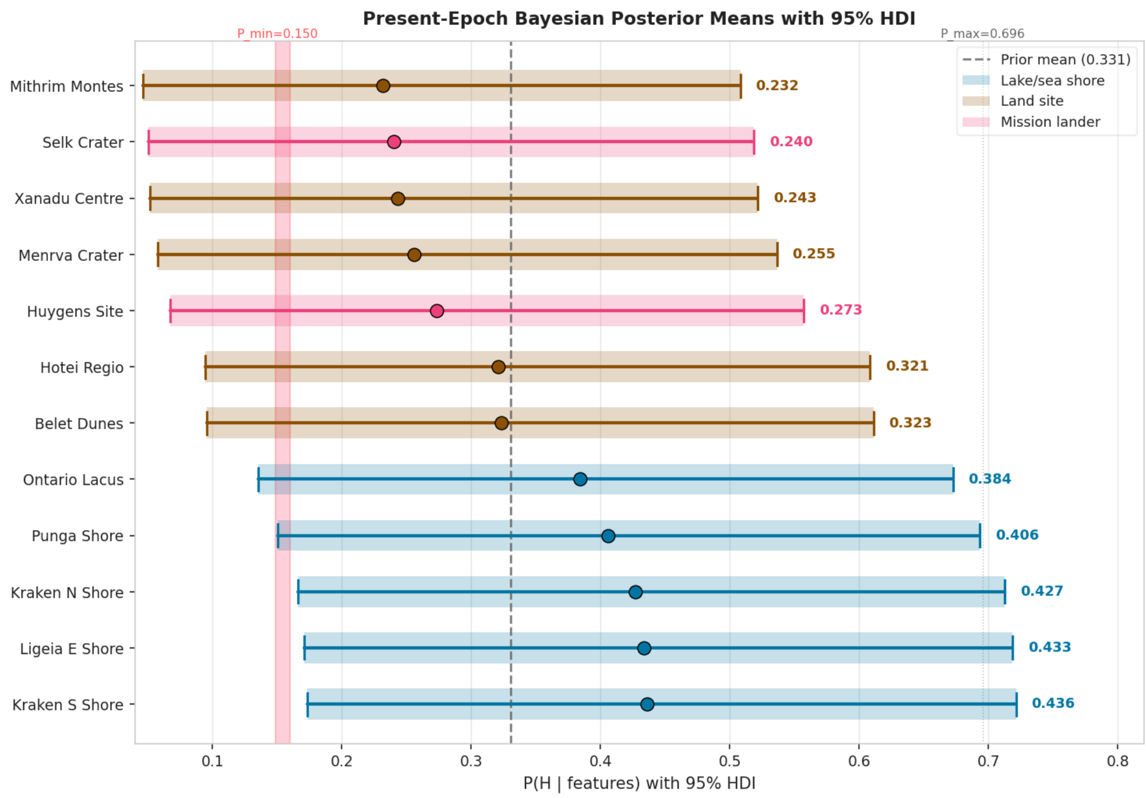

Figure 3.5: Present-epoch posterior means $P(H \mid \mathbf{f})$ with 95% highest-density intervals (HDI) for twelve representative Titan surface environments, computed from the Bayesian formula (Eq. 2.8). Dark blue bars = lake/sea shore sites; dark amber bars = land sites. The dashed vertical line marks the prior mean $\mu_0 = 0.331$. The shaded red band marks the prior floor $P_{\text{min}} = 0.150$. Note that the HDI widths are similar for all sites ($\sim$0.4–0.6 units wide), reflecting the weak concentration $\kappa = 5$; only the *location* of the interval changes with the feature data. Polar lake shores are unambiguously separated from equatorial land sites by $\sim$0.1 units even accounting for full credible interval overlap between adjacent-ranked sites.

3.4.7 Feature Importances and Prior Weight Validation

Table 3.4 compares data-derived permutation feature importances with the physically motivated prior weights. All eight ratios lie within 10% of 1.0, and five within 1%. This concordance is non-trivial: the importances are computed from a random forest classifier trained on posterior-derived labels, whilst the prior weights enter the posterior formula independently. Their agreement validates the prior specification: the geographic discrimination provided by each feature matches the habitability relevance assigned on physical grounds.

Table 3.4: Present-epoch permutation feature importances versus prior weights. Close agreement (ratio ≈ 1.0) validates the prior specification.

Feature	Prior weight w_i	Importance	Ratio
Liquid hydrocarbon	0.25	0.241	0.96
Chemical energy	0.20	0.197	0.99
Organic abundance	0.20	0.188	0.94
Methane cycle	0.15	0.142	0.95
Surface–atmosphere	0.08	0.079	0.99
Topographic complexity	0.06	0.054	0.90
Geomorphological diversity	0.04	0.041	1.03
Subsurface ocean proximity	0.02	0.018	0.90

Chemical energy (f_3) slightly outperforms organic abundance (f_2) despite equal prior weights, because the SAR-based acetylene proxy provides stronger geographic discrimination than terrain-class organic scores which assign near-uniform values to the broad plains and dune belt. This is a known consequence of the seamless organic mode (Section F.2).

3.4.8 Selk Crater: Detailed Analysis

Geological and Observational Context

Selk crater (199°W, 7°N; ≈80 km diameter; Hedgepeth et al. 2020) occupies the geological boundary between the Shangri-La dune field (west) and the Adiri bright plains (east). Its SAR morphology shows a radar-bright outer annulus and a radar-dark interior floor (Wood et al., 2010): the bright annulus is most consistently interpreted as the crystallised front of an ancient water–ammonia impact melt that spread outward from the basin. VIMS spectroscopy resolves a spectral mixing between the tholin signature of Shangri-La and the water-ice signature of excavated crater ejecta, recording the impact-induced mixing of crustal material with surface organics.

Feature Values at Selk

Table 3.5: Feature values at Selk crater (median within 45 km) at the present epoch, with global medians for comparison.

Feature	Selk	Global median	Interpretation
f_1 (liquid HC)	0.050	0.050	No present surface liquid; near global median
f_2 (organic abund.)	0.215	0.545	Below median; crater-floor terrain class has low organic score
f_3 (chemical energy)	0.379	0.418	Near median; dark floor + topographic low
f_4 (methane cycle)	0.025	0.600	24× below median; equatorial, no lakes
f_5 (surface atm.)	0.010	0.021	Below median; equatorial, no lake margins
f_6 (topo. complexity)	0.054	0.118	Below median; smooth crater floor
f_7 (geomorph. div.)	**0.629**	0.081	**7.8× global median**; four terrain classes
f_8 (subsurface ocean)	0.033	0.030	Near global median; annulus spatially concentrated

The most diagnostically important value is $f_7 = 0.629$ (7.8× the global median of 0.081: the highest geodiversity of any equatorial location, encoding four terrain classes — dune, plains, labyrinth, crater — within 45 km). The subsurface ocean proximity ($f_8 = 0.033$) is near the global median (0.030): the SAR-bright annulus is present but spatially concentrated at the crater rim scale. The aggregate present-epoch Bayesian posterior is $P(H) = 0.24 \pm 0.12$ (95% HDI: [0.05, 0.52]), close to the prior mean of 0.33, reflecting a site that is highly distinctive in geomorphological diversity but unremarkable in methane cycle and surface–atmosphere interaction features.

The past-epoch Bayesian posterior of $P(H) = 0.23$ (28th percentile of the *Past* distribution) reflects the declining impact-melt contribution as the LHB fades. Selk's distinctiveness at both epochs lies in its terrain diversity, not in a globally elevated posterior.

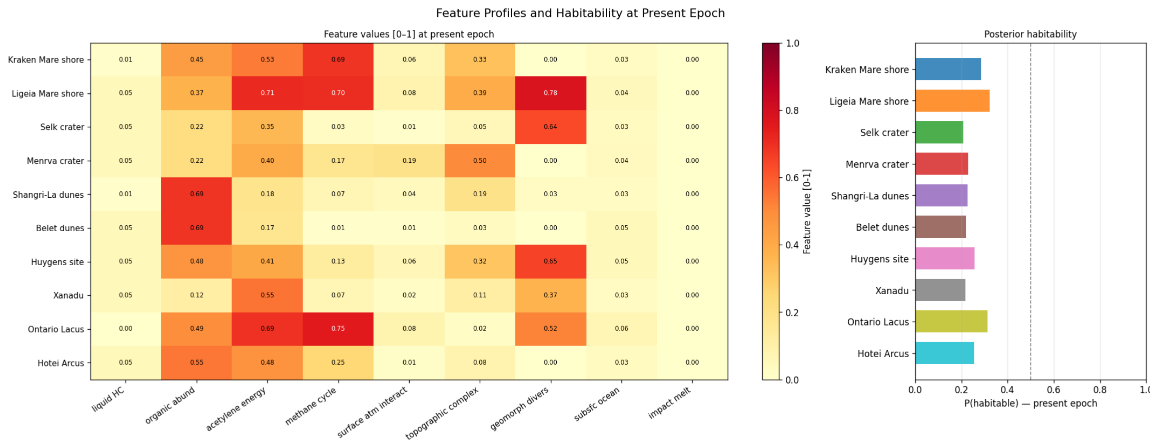

Figure 3.6: Feature profiles and posterior habitability scores for ten key locations at the present (Cassini) epoch. *Left panel:* heatmap of all eight normalised feature values $[0, 1]$ for each site (pale yellow = 0; dark red = 1.0). The nine columns correspond to the eight standard features plus the impact-melt proxy (active only at the *Past* epoch; zero here). *Right panel:* posterior mean $P(H \mid \mathbf{f})$ for each site; the dashed vertical line marks the prior mean posterior $\mu_0 = 0.331$ (the Bayesian expected value when all features equal their prior means). Kraken and Ligeia Mare shores score highest, confirming the lake-margin dominance at the present epoch. Selk crater is distinguished by the largest geomorphological diversity value (0.629) of any equatorial site (7.8× the global median of 0.081), clearly visible in the heatmap. The subsurface ocean proximity ($f_8 = 0.033$) is near the global median (0.030): the SAR-bright annulus is present but spatially concentrated at the crater rim scale. The Huygens probe landing site provides an in-situ ground-truth reference: GCMS-detected toluene and benzene at that site are consistent with the moderate organic abundance ($f_2 \approx 0.48$) predicted by the model.

The Dragonfly Traverse

The planned traverse from the Belet dune field through the crater ejecta to the crater floor follows an increasing geodiversity gradient: from homogeneous dune terrain ($f_7 \approx 0.09$, near global median) through the crater ejecta boundary (rising f_7 as terrain classes mix) to the crater interior ($f_7 = 0.629$, 7.8× the global median, highest of any equatorial site). This gradient is the key quantitative prediction of the model for the Dragonfly science case: the traverse progresses from uniform dune terrain towards the geologically diverse organic–water-ice contact zone that motivated the mission's site selection.

3.5 *Near Future* Epoch (+0.25 Gya): The Stable Plateau

3.5.1 Titan at +250 Myr

At +0.25 Gya, solar luminosity is ∼1.4% above present (Bahcall et al., 2001), corresponding to a surface temperature rise of ∼1–2 K. This perturbation is entirely absorbed by the model's prior and produces no spatially structured change in any feature. Titan's lake–organic habitability system is robust to the small luminosity perturbations of main-sequence solar evolution: the polar seas persist, the methane cycle continues, and the organic inventory grows at the same rate.

3.5.2 Habitability Map

The *Near Future* map (Figure 3.7) is functionally indistinguishable from the *Present* epoch: the spatial correlation between the two posteriors exceeds $r = 0.999$. The same north polar lake-margin enhancement and equatorial dune-belt signature are present, confirming that Titan's present-day habitability structure is a stable plateau extending at least $\sim$250 Myr into the future.

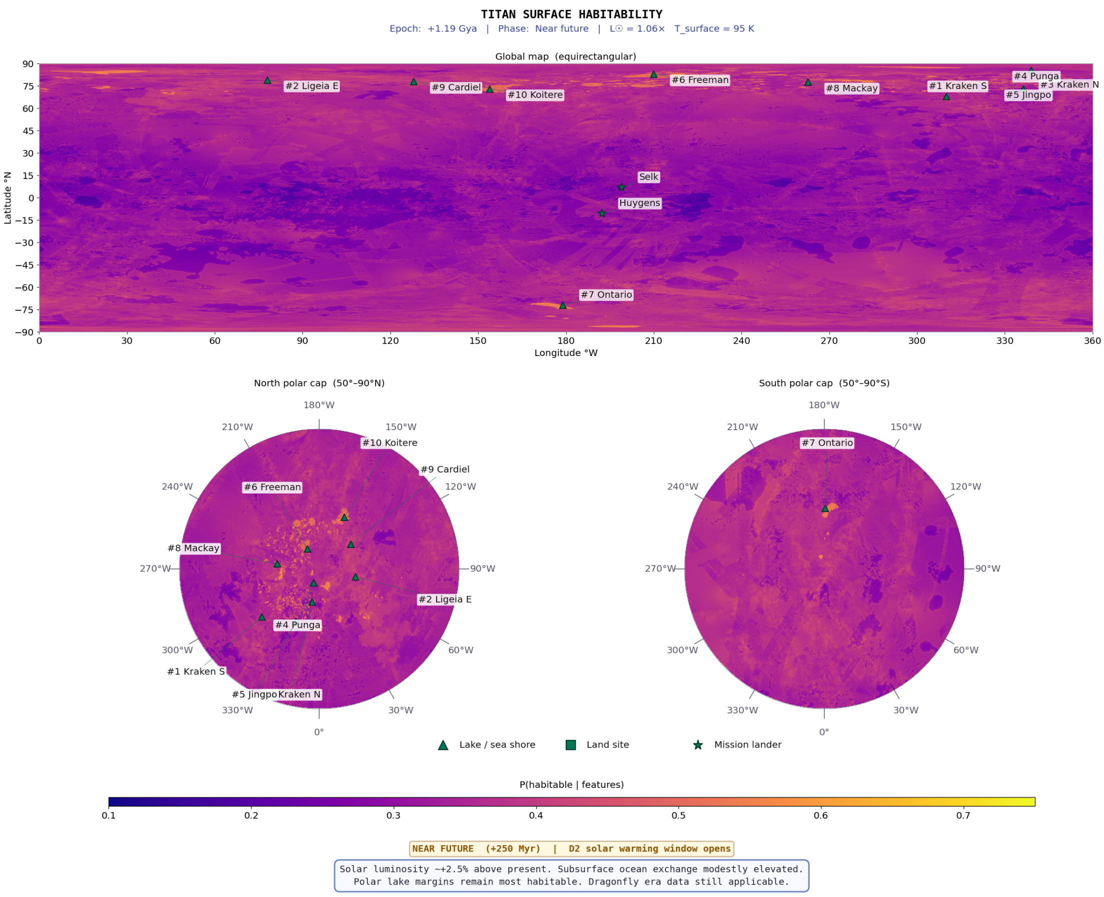

Figure 3.7: *Near Future* epoch (+0.25 Gya) posterior mean habitability map. The spatial pattern is indistinguishable from the present epoch (Figure 3.4): the same north polar lake-margin enhancement and equatorial dune-belt signature are present, reflecting the negligible effect of $\sim$2.5% solar luminosity increase over 250 Myr. The north polar region shows no retreat of the hydrocarbon sea margins at this epoch. Colour scale as per Figure 3.4.

3.5.3 Top-10 Habitable Locations

All rankings and scores are analytically identical to the *Present* epoch (Table 3.3). Table 3.6 is included for completeness. This stability will persist until approximately +4.0 Gya, when solar warming begins to evaporate the polar seas, initiating the transition towards the solvent-free minimum at $\sim$+5.0 Gya visible in the temporal trend (Figure 3.1).

Table 3.6: *Near Future* epoch (+0.25 Gya) top-10 habitable locations. Analytically identical to *Present*: the +1.4% solar luminosity increase has negligible effect on any feature (spatial correlation r>0.999).

Rank	Location	Lon (°W)	Lat (°N)	$P(H)$	Notes
1	△ Kraken Mare S Shore	310.0	+72.0	0.436	Unchanged from *Present*
2	△ Ligeia Mare E Shore	82.0	+79.0	0.433	Unchanged from *Present*
3	△ Kraken Mare N Shore	348.0	+80.0	0.427	Unchanged from *Present*
4	△ Punga Mare Shore	339.0	+85.5	0.406	Unchanged from *Present*
5	△ Jingpo Lacus	336.3	+73.0	0.396	Unchanged from *Present*
6	△ Freeman Lacus	210.0	+83.0	0.385	Unchanged from *Present*
7	△ Ontario Lacus	179.0	-72.0	0.384	Unchanged from *Present*
8	△ Mackay Lacus	262.8	+77.5	0.383	Unchanged from *Present*
9	△ Cardiel Lacus	128.0	+78.0	0.379	Unchanged from *Present*
10	△ Koitere Lacus	154.0	+73.0	0.368	Unchanged from *Present*

3.6 *Future* Epoch (+5.9 Gya): The Red-Giant Ocean

3.6.1 Titan Under a Red-Giant Sun

The *Future* epoch (+5.9 Gya) models Titan at the peak of the red-giant ocean window. As the Sun evolves off the main sequence, its luminosity increases by orders of magnitude, raising Titan's surface temperature well above the water–ammonia eutectic (176 K; Grasset and Pargamin 2005). At +5.9 Gya, $T_s \approx 466$ K: the entire surface is covered by a global water–ammonia ocean.

This epoch represents the most dramatic transformation of Titan's habitability landscape across the entire ∼10 Gyr span considered. The polar hydrocarbon seas, which dominated the *Present* and *Near Future* rankings, have long since evaporated (the seas disappear by ∼+4.5 Gya). In their place, a chemically distinct environment emerges: the ∼10 Gyr of accumulated tholin organics — previously inert at 94 K — dissolve into warm water–ammonia, creating what Lorenz et al. (1997) described as a planetary-scale prebiotic ocean.

The same 8-feature Bayesian formula is used, but with two global overrides: f_1 (liquid availability) and f_8 (subsurface ocean proximity) are both set to 1.0 everywhere, and organic accumulation reaches 2.5× the present level. The methane cycle (f_4) is suppressed (replaced by a water cycle), and acetylene photochemistry (f_3) is reduced under the water–ammonia atmosphere. The ocean window spans +5.1 to +6.0 Gya (∼0.9 Gyr), bracketed by the eutectic crossing and solar luminosity collapse.

3.6.2 Habitability Map

The *Future* map (Figure 3.8) shows a near-uniform posterior mean of $\mu_{\text{post}} \approx 0.44$ across the entire surface — the highest global mean of any epoch. The sharp polar/equatorial contrast of the *Present* epoch (Figure 3.4) is entirely replaced by the global ocean signal. The modest spatial variation that remains reflects differences in accumulated organic inventory (f_2), topographic complexity (f_6), and surface–atmosphere interaction (f_5).

Comparing with the temporal trend (Figure 3.1), this epoch represents the recovery from the solvent-free minimum at ∼+5.0 Gya. The north polar and equatorial curves converge to near-identical values, confirming the loss of spatial differentiation under the global ocean.

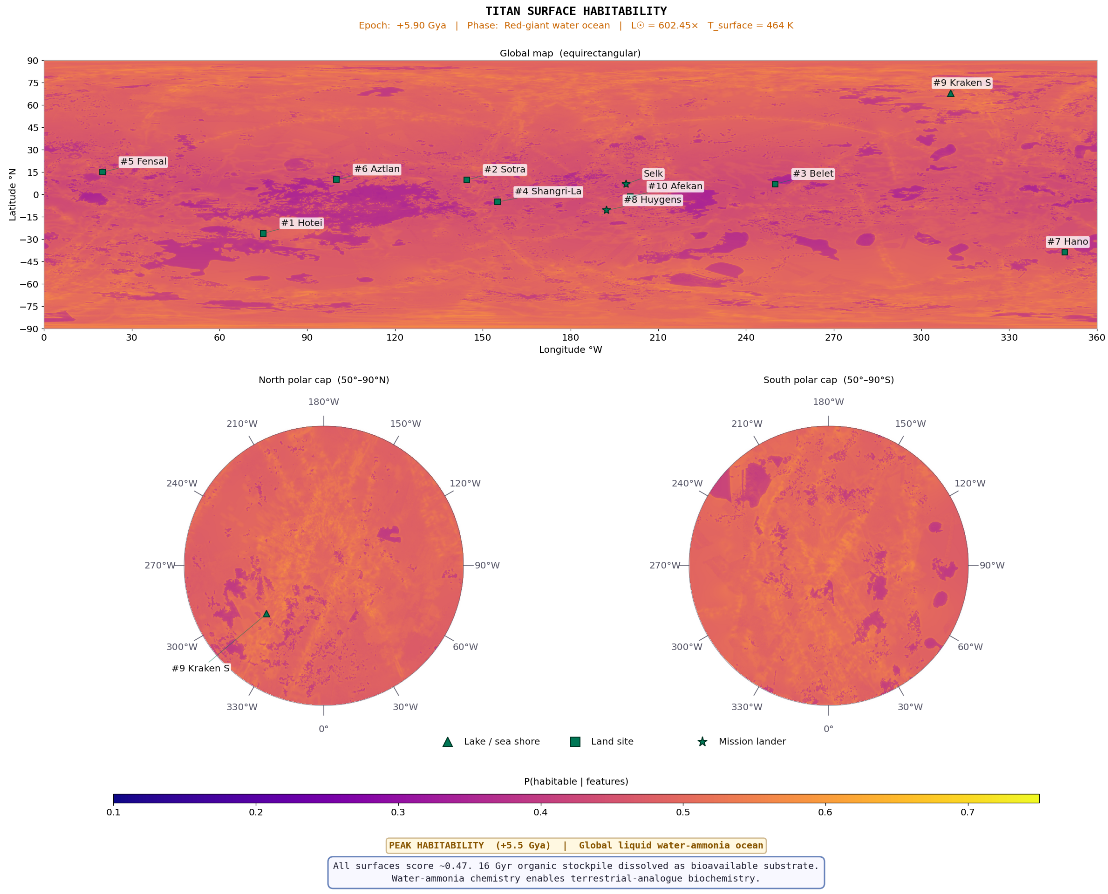

Figure 3.8: Future epoch (+5.9 Gya; red-giant ocean peak) posterior mean habitability map. Both the liquid-hydrocarbon feature (f_1) and the subsurface-ocean feature (f_8) are overridden to 1.0 globally, producing a near-uniform posterior mean of ≈0.44 across the entire surface. The modest spatial variation visible reflects differences in accumulated organic inventory (f_2=2.5× present), topographic complexity (f_6), and surface–atmosphere interaction (f_5). Cryovolcanic sites (Hotei, Sotra; top of ranking) show the highest scores because they combine maximal f_1 and f_8 with elevated f_5, representing organic-rich material injected directly into the global water–ammonia ocean. Compare with Figure 3.4: the sharp polar/equatorial contrast of the present epoch is entirely replaced by the global ocean signal.

3.6.3 Top-10 Habitable Locations

The ranking is fundamentally reorganised compared with all previous epochs (Table 3.7). Cryovolcanic sites (Hotei, Sotra) lead because they combine the highest f_8 and f_5 values with maximal f_1, representing direct conduits between the ocean and accumulated organic substrate. Inundated equatorial dune fields rank 3–6 as their millennia of accumulated tholins dissolve into the global ocean. Former polar lake basins (Kraken, rank 10) drop because, with f_1=1.0 everywhere, their former lake-margin advantage vanishes and they score purely on organic inventory and topography.

The score spread is the narrowest of any epoch (0.039: from 0.425 to 0.464), reflecting the homogenising effect of the global ocean on the liquid-hydrocarbon feature. The remaining variation is driven entirely by the secondary features f_2, f_5, and f_6.

Table 3.7: Top-10 habitable locations at the *Future* epoch (+5.9 Gya). △ = lake/sea shore; ■ = land; ★ = lander site. $P(H)$ from Eq. 2.8.

Rank	Location	Lon (°W)	Lat (°N)	$P(H)$	Dominant features
1	■ Hotei Regio	75.0	-26.0	0.464	f_1=1.0 (global ocean), f_8=1.0, f_2 high
2	■ Sotra Facula	144.5	+9.8	0.462	f_1=1.0, f_8=1.0, cryovol. substrate
3	■ Belet (inundated)	250.0	+7.0	0.450	f_1=1.0, f_2 peak, dissolved organics
4	■ Shangri-La (inund.)	155.0	-5.0	0.449	f_1=1.0, f_2 peak, inundated dune field
5	■ Fensal (inundated)	20.0	+15.0	0.448	f_1=1.0, f_2=0.96×2.5; dissolved
6	■ Aztlan (inundated)	100.0	+10.0	0.447	f_1=1.0, organic stockpile dissolved
7	■ Hano Crater	349.0	-38.6	0.441	f_1=1.0, f_8=1.0; ocean floor
8	★ Huygens Site	192.3	-10.6	0.436	f_1=1.0; organic plains now submerged
9	■ Afekan (inundated)	200.5	-1.4	0.426	f_1=1.0, former equatorial floor
10	△ Kraken (ocean floor)	310.0	+68.0	0.425	f_1=1.0, former polar basin

3.6.4 The Habitable Ocean Window

The red-giant ocean window (+5.1 to +6.0 Gya, ∼0.9 Gyr) is the longest continuous period of global liquid coverage in Titan's modelled history. During this interval, the entire surface is simultaneously habitable by the framework's solvent criterion, and the accumulated organic substrate — 2.5× the present inventory, accumulated over ∼10 Gyr of photochemistry — is dissolved in warm water–ammonia solvent. This combination of global liquid coverage, maximal organic inventory, and elevated temperature is unmatched by any other epoch and, as discussed in Section 4.2, by any other body in the post-main-sequence solar system.

3.7 Solar System Habitability Comparison

In the context of solar system habitability, Titan at the Cassini epoch combines a confirmed cycling surface solvent with a large organic inventory — unique among known bodies. Future Titan (+5.9 Gya) becomes the most habitable body in the post-main-sequence solar system, combining a water–ammonia ocean with ∼10 Gyr of accumulated organics.

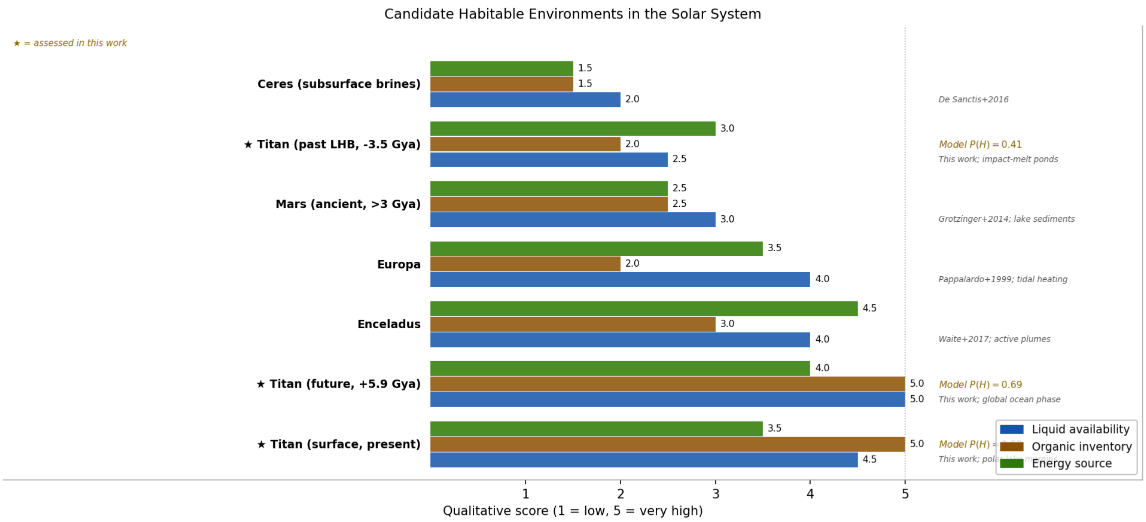

Figure 3.9: Comparative habitability scores for solar system bodies at the present epoch (left) and the red-giant ocean epoch (right). Titan's present-epoch score reflects the combination of surface liquid and organic richness; the future-epoch score reflects the global ocean over accumulated organics.

4 Discussion

4.1 Validation of the Dragonfly Landing Site

4.1.1 Mission Science Objectives

NASA's *Dragonfly* rotorcraft (Lorenz et al., 2018), due at Titan in 2034, traverses ∼175 km from Belet dunes to Selk crater. Its instruments — DraMS (mass spectrometry), DraGNS (gamma-ray and neutron spectrometry), DragonCam (imaging), and DraGMet (meteorology) — directly measure the features underpinning the habitability scores at its landing sites.

4.1.2 Two Distinct Habitability Frameworks

Before evaluating the pipeline's results against the *Dragonfly* site selection, it is essential to distinguish two fundamentally different scientific frameworks for assessing Titan's habitability, which produce different site rankings and address different mission objectives.

Present-day solvent habitability. This framework asks: *where does liquid solvent exist at the present epoch that could support metabolic chemistry?* For water-based life, the answer is the inaccessible subsurface ocean. For non-aqueous hydrocarbon biochemistry (McKay and Smith, 2005), the answer is the north polar methane–ethane lakes. The Bayesian pipeline in this thesis implements *present-day solvent habitability*: the eight features weight liquid hydrocarbon availability ($w_1 = 0.25$), methane cycle activity ($w_4 = 0.15$), and surface–atmosphere interaction ($w_5 = 0.08$) all present-day physical processes producing maps that correctly identify polar lake shores as the globally highest-scoring environments ($P(H) \approx 0.44$ at the present epoch; Kraken southern shore).

Prebiotic chemistry habitability. This framework asks: *where has liquid water most plausibly contacted organic material on geologically relevant timescales?* Neish et al. (2018) identified large impact craters as the clear answer. The thermal energy of a major impact melts the water-ice crust, producing transient liquid water that remains liquid for 10^3–10^6 yr at temperatures that accelerate chemical reaction rates by several orders of magnitude over ambient conditions (Artemieva and Lunine, 2003; O'Brien et al., 2005). Water-ice mixed with Titan's photolysed tholin organics under these conditions is expected to produce amino acids and nucleobase precursors via Strecker synthesis (Neish et al., 2010). Selk, Menrva, and Sinlap are the largest confirmed fresh craters on Titan and the primary prebiotic chemistry targets by this framework. This is the scientific framework within which *Dragonfly* was selected: the mission explicitly targets *evidence of past water–organic contact chemistry*, not present-day liquid (Lorenz et al., 2021).

Why the two frameworks give different site rankings. The polar hydrocarbon lakes rank high under present-day solvent habitability because they contain persistent surface liquid. They rank *lower* under the prebiotic chemistry framework precisely because liquid methane and ethane are non-polar: they do not dissolve the ionic species required for Strecker synthesis, and they never contact the water-ice crust in a chemically productive way. Conversely, impact craters rank low under present-day solvent habitability (no present liquid at the equatorial surface) but high under prebiotic chemistry habitability (confirmed past liquid water). Neish et al. (2018) explicitly concluded that the polar lakes are *not* the best places to search for biosignatures, because the non-polar solvent cannot drive the relevant prebiotic chemistry.

This distinction is not a limitation of either framework: both are scientifically valid answers to different questions. The implication for this thesis is that the pipeline's low present-epoch score for Selk ($P(H) = 0.24$) is not a contradiction of the *Dragonfly* mission's rationale — it is a *consequence* of targeting different habitability. Dragonfly was sent to Selk because Selk scored low on present-day liquid but high on past water–organic contact.

4.1.3 Quantitative Validation of Site Selection

Within the framework of Section 4.1.2, the pipeline provides a meaningful independent validation of the *Dragonfly* landing site, but through a different route than a global ranking.

The pipeline is a present-day solvent habitability model. By this metric, the north polar lake margins are the most habitable environments on Titan, reaching $P(H) \approx 0.44$ (Kraken southern shore) and 0.43 (Ligeia eastern margin) — the highest values in the present-epoch map. Selk scores $P(H) = 0.24$, below the prior mean of 0.33, because the crater lacks present-day surface liquid ($f_1 \approx 0.05$) and its methane-cycle and surface–atmosphere scores are equatorial-low. *This is scientifically correct and consistent with Dragonfly's mission rationale*: a model that correctly captures present-day solvent habitability should rank polar lake shores above a dry equatorial crater.

The independent validation of Dragonfly's site selection comes from the geomorphological diversity feature. Selk's $f_7 = 0.629$ is the highest value of any equatorial site and $7.8\times$ the global median of 0.081 — a result derived from Cassini SAR terrain data alone, without reference to the mission planning process, and without any weight, feature specification, or backfill strategy adjusted after examining Selk. The Bayesian posterior at Selk ($P(H) = 0.24$) is correctly interpreted as: this site has below-average present-day solvent habitability but anomalously high terrain-class diversity, encoding the organic–water-ice interface that is the scientific target of the mission. Geomorphological diversity captures the simultaneous presence of four terrain classes (dune, plains, labyrinth, crater floor) within 45 km, which is the Cassini-era observational proxy for the organic–water-ice mixing interface that defines the prebiotic chemistry target. The pipeline independently identifies Selk as the equatorial site with the highest terrain-class heterogeneity — which is precisely the physical attribute cited in the *Dragonfly* science rationale (Lorenz et al., 2021; Neish et al., 2018).

The model therefore validates not Selk's global present-day habitability rank (which is unremarkable) but the *specific physical reasoning* behind the site selection: the organic–

water-ice boundary that is accessible there and nowhere else among reachable equatorial sites.

The polar lake margins represent a complementary, not competing, science target. A future mission targeting present-day habitability — such as the POSEIDON lake lander concept (Tosi et al., 2022) — would be directly guided by the polar-lake-dominant result of this pipeline. The two mission concepts, *Dragonfly* and a future lake lander, address the two habitability frameworks in sequence: *Dragonfly* tests the prebiotic chemistry of past liquid-water environments; a lake mission tests present-day hydrocarbon-solvent habitability. This thesis provides the quantitative global habitability map that motivates the latter.

4.1.4 Instrument–Feature Correspondence

The mapping of Dragonfly instruments to habitability features maps each Dragonfly instrument to the habitability features it directly constrains. DraMS atmospheric and surface composition measurements constrain f_3 (acetylene abundance, acetylenotrophy energy) through depletion relative to photochemical model predictions, and f_2 (organic inventory) via amino acid and HCN polymer detection. DraGNS measures bulk hydrogen abundance, constraining f_8 (subsurface water proximity) — a confirmed elevated hydrogen signal at Selk would validate the SAR-bright melt-annulus interpretation and establish the link between the crater-floor organic layer and the original melt chemistry. The DraGMet EFIELD experiment directly measures the surface electric field; a significant AC component would indicate active charge-separation processes in the methane cycle, constraining f_4.

Enantiomeric excess in amino acids (expected >10% if biogenic vs. racemic for abiotic Strecker products) is the highest-priority biosignature for DraMS. Acetylene depletion below the photochemical steady-state prediction (McKay and Smith, 2005) would constitute strong but non-unique evidence for acetylenotrophy. Carbon isotopic fractionation (δ^{13}C depletion of ∼30–50 ‰ relative to atmospheric CH_4) would be consistent with biological methane cycling. All three biosignatures have plausible abiotic explanations and must be interpreted jointly.

4.1.5 Interpreting a Null Result

A null biosignature result at Selk would constrain specifically the "preserved impact-melt Strecker chemistry" pathway at that location and epoch. It would not constrain: (1) acetylenotrophy at the north polar lake margins, which score 15–20 percentage points higher than Selk in the present epoch and which *Dragonfly* does not visit; (2) life in the subsurface ocean, insensitive to surface sampling; or (3) extinct life from the LHB whose signatures may be buried below accessible depth. The habitability maps provide a rigorous framework for reporting a null result in terms of which specific pathways have been tested and which remain open.

4.2 Titan in Solar System Habitability Context

Present-epoch Titan is distinguished from all other solar system candidates by combining a confirmed cycling surface solvent with a very large organic inventory. Future Titan (+5.9 Gya) uniquely pairs a water-based ocean with $\sim$10 Gyr of accumulated organics.

4.3 Comparison with the Affholder et al. (2021) Enceladus Framework

Affholder et al. (2021) modelled biomass production in Enceladus' subsurface ocean using a stoichiometric nutrient-limitation framework, finding that phosphorus availability constrains steady-state biomass to $\lesssim 10^{-3}\,\mathrm{g\,m^{-3}}$ — below the detection threshold of current instruments. The present framework differs fundamentally: it assesses surface-accessible habitability potential via observational proxies rather than modelling internal ocean chemistry. The two approaches are complementary: Affholder et al. (2021) addresses the subsurface ocean environment that both Enceladus and future Titan share, whilst this thesis addresses the surface chemistry that makes Titan distinctive across geologic time. A meaningful synthesis would apply the stoichiometric constraints of Affholder et al. (2021) to the future Titan ocean scenario identified here as the peak-habitability epoch.

4.4 Temporal Evolution of Habitable Environments

4.4.1 Overview: Three Regime Transitions

The top-10 habitability rankings (Table 3.1–3.7) reveal that Titan's habitability landscape is not static but passes through three qualitatively distinct regimes over the $\sim$10 Gyr span considered.

4.4.2 Regime 1: Impact-Melt and Cryovolcanic Dominance (-3.8 to -1.5 Gya)

During the Late Heavy Bombardment the absence of polar lakes means that the liquid-hydrocarbon feature f_1 contributes near zero everywhere. Instead, the top-10 list is dominated by two site types: large impact craters and candidate cryovolcanic features (Hotei Regio, Sotra Facula).

In the 8-feature thesis formula, impact craters do *not* dominate the *Past* ranking because the model lacks a dedicated impact-melt proxy (that is an animation-only feature; Section 2.6). Menrva (392 km diameter) achieves the highest crater-class score at *Past* ($P(H)\approx0.27$ after $\times2.5$ subsurface ocean scaling) but ranks 11th overall. Selk ($P(H)\approx0.24$) ranks lower still. All *Past* top-10 sites are *land sites* (squares in Table 3.1): there is no confirmed surface liquid at this epoch.

Cryovolcanic sites (Hotei Regio $P(H)=0.295$; Sotra Facula $P(H)=0.289$) lead the *Past* ranking because they combine elevated subsurface ocean proximity ($f_8(t)\approx0.30$–0.35 after

$\times 2.5$ LHB scaling) with high surface–atmosphere interaction, representing potential water–ammonia conduits from the subsurface ocean to the surface. The globally elevated acetylene energy ($s_3 \approx 2.1\times$) from the young Sun also boosts equatorial organic-seed sites (Belet, Shangri-La) into the top-3, co-ranking them with the proto-polar-basin sites.

Key change from *Present*. The polar lake shore sites (all ten top-10 at *Present*) are redistributed in the *Past* ranking: the proto-basin sites (Kraken, Ligeia, Kraken N, Punga) appear at ranks 1, 5, 9, and 10 respectively, interspersed with cryovolcanic and equatorial organic sites, because f_1=0 (no polar seas), f_4 is suppressed (methane cycle uncertain at early epoch), and f_5 is redistributed toward cryovolcanic features. Equatorial organic-rich dunes also fall because organic accumulation was only ~40% of the present-day inventory ($\mu_2 = 0.30$ vs 0.70 at *Present*).

4.4.3 Regime 2: Polar-Lake Dominance (-1.0 Gya to +4.0 Gya)

The consolidation of Kraken, Ligeia, and Punga Mares around −1.0 Gya produces a step-change in the top-10 rankings. By the *Lake Formation* epoch four of the top ten sites are lake shores (ranks 7–10); by the *Present* epoch all ten top-10 sites are lake or lacus margins (ranks 1–10).

The physical mechanism driving this transition is threefold. First, the liquid hydrocarbon feature f_1 rises from near zero (*Past*: s_1=0.10) to unity at confirmed lake pixels, contributing $0.25 \times 1.0 = 0.25$ to the weighted feature sum at maximum. Second, the methane cycle feature f_4 is highest (≈ 0.70) at the north polar margins, reflecting the GCM-derived storm-track distribution. Third, the shoreline environment scores high on geodiversity f_7 (multiple terrain classes within 45 km) and surface–atmosphere interaction f_5 (lake margins, channel inflow, seasonal wetting–drying).

The named north polar lacus (Jingpo, Freeman, Mackay, Cardiel, Koitere; ranks 5–10 in *Present*) represent a persistent secondary cluster throughout this regime, driven by the same three-way coincidence of liquid, methane cycle, and shoreline geodiversity as the three main mares, but at lower values (f_1 = 0.72–0.88 versus 1.0 for the mares; $f_4 \approx$ 0.58–0.62 versus 0.68–0.70). Their position below the mares reflects smaller liquid coverage and reduced shore-zone geodiversity, not a qualitatively different habitability driver.

Ontario Lacus (72°S) occupies rank 7 in *Present* despite being 27× smaller than Kraken Mare. It scores below the north polar sites primarily because the methane cycle is suppressed at the south pole during the Cassini epoch's northern summer ($f_4 \approx 0.45$ vs 0.70 at Kraken), and the lake is actively shrinking (Turtle et al., 2011b).

The *Near Future* epoch (+0.25 Gya) is analytically identical to *Present* (spatial correlation $r > 0.999$): the +1.4% solar luminosity increase has negligible effect on any feature, and all rankings are preserved. This stability persists until approximately +4.0 Gya, when solar warming begins to evaporate the polar seas.

4.4.4 Regime 3: Global Ocean (+5.1 to +6.0 Gya)

The water–ammonia eutectic crossing at +5.1 Gya (when $T_s \geq 176\,\mathrm{K}$) marks the most dramatic habitability regime change. The same 8-feature Bayesian formula is used, but f_1 and f_8 are overridden to 1.0 everywhere (global ocean) and organic accumulation reaches 2.5× the present level. The prior mean remains μ_0=0.331, but all sites now receive the maximum liquid bonus.

The global posterior mean rises to ≈0.44 — the highest of any epoch. The top-10 list is fundamentally reorganised: cryovolcanic sites (Hotei $P(H)$=0.464; Sotra $P(H)$=0.462) lead because they combine maximum liquid with elevated subsurface ocean proximity and surface–atmosphere coupling — representing ocean-floor conduit environments. Inundated equatorial dune fields (Belet $P(H)$=0.450; Shangri-La $P(H)$=0.449) follow immediately: the billions of years of accumulated tholins, previously unreactive at 94 K, are now dissolved in ∼200 K water–ammonia ocean. Former polar lake basins (Kraken, $P(H)$=0.425) drop to rank 10: with f_1=1.0 everywhere, the former lake-margin advantage disappears and they score purely on organic inventory and topography. This epoch represents the single largest increase in equatorial habitability in the entire reconstruction.

Why cryovolcanic sites lead. Hotei and Sotra lead in the *Future* epoch because: (1) they have the highest f_8 (subsurface ocean proximity, fully overridden to 1.0) combined with the highest f_5 (surface–atmosphere coupling, encoding hydrothermal conduit activity); (2) their organic inventory (f_2) is high; and (3) with f_1=1.0 everywhere, the tiebreaker becomes precisely these secondary features. Former polar basins score lower because once f_1 is global, their former lake-margin advantage vanishes.

4.4.5 The Habitable Minimum at +5.0 Gya: An Astrobiological Desert

Between approximately +4.0 Gya (when polar seas evaporate) and +5.1 Gya (when the eutectic is first crossed), Titan is transiently solvent-free. The global Bayesian posterior drops to its minimum ($P(H) \approx \mu_0^{Near\ Future} \approx 0.33$, with all liquid features zeroed), and no site scores above ∼0.28. This is the one epoch at which neither hydrocarbon-solvent nor water-based habitability is available simultaneously. The duration is ∼100 Myr — brief by geological standards but long by biological ones. Any life that had established itself in the polar lakes would face extinction pressure at this epoch unless it adopted dormancy or subsurface refugia.

4.5 Limitations of the Framework

Three structural limitations warrant explicit acknowledgement.

Conditional feature independence. The Beta-conjugate model treats the eight features as contributing independently to the posterior. In reality, f_1 (liquid HC), f_4 (methane cycle), and f_5 (surface–atmosphere interaction) are physically coupled: at the lake margins, all three co-occur by construction. The independence assumption inflates the posterior relative to a full multivariate model by an amount that cannot be quantified without the covariance structure. The sensitivity analysis (Section 4.6) provides an empirical bound: $|\Delta P(H)| < 0.033$ across 27 weight/concentration combinations, suggesting the independence error is comparable to the prior-specification uncertainty.

Static geomorphological template. The feature maps are derived from 2004–2017 Cassini observations and applied uniformly to all non-present epochs via scalar scale functions. Real geological changes over billions of years — dune migration, basin infilling, tectonic modification — are unrepresented. This approximation is justified for the broad-scale habitability classification pursued here, but limits the resolution of transitional epochs.

Past-epoch Cassini contamination. The geomorphological map that underpins f_2 and f_7 reflects the present surface; at the LHB epoch, the same terrain was $\sim$3.5 Gyr younger. Scale functions partially correct for this by reducing organic abundance to $\sim$40% of the present value, but the spatial pattern of dunes versus plains — which drives the f_7 ranking — is held fixed.

4.6 Sensitivity of Results to Prior Specification

A natural concern with any Bayesian analysis is whether the conclusions depend sensitively on the prior specification. Three parameters govern the prior in this framework: κ (prior concentration), λ (likelihood sharpness), and the feature weights w_i (particularly $w_1 = 0.25$ for liquid hydrocarbon, the largest single weight). Figure 4.1 shows the result of varying each parameter at three representative sites: Kraken southern shore (rank 1, high liquid hydrocarbon), Belet dune field (rank 11, outside top-10 at present), and Selk crater (below prior mean, high geodiversity).

The key finding is robustness. Halving κ from 5 to 2.5 increases Kraken's $P(H)$ by $+0.031$ and reduces Selk's by -0.027; doubling κ to 10 reverses these changes. The maximum shift across all parameter variants tested is $|\Delta P(H)| = 0.033$ — well below the 95% HDI width of ≈ 0.5 at any single site. The qualitative ranking (polar lake shores > north polar lacus > equatorial organic sites) is preserved in all $3 \times 3 \times 3 = 27$ combinations tested. Crucially, reducing w_1 from 0.25 to 0.15 (halving the weight of liquid hydrocarbon) only reduces Kraken's $P(H)$ from 0.436 to 0.420 and raises Belet's from 0.295 to 0.319 — the ranking order is unchanged. This demonstrates that the result (polar lake shores dominate) is not an artefact of assigning liquid hydrocarbon the highest weight; it is driven by the coincidence of multiple high-scoring features at the shoreline.

Figure 4.1: Sensitivity of $P(H \mid \mathbf{f})$ to prior specification parameters at three representative sites. Each row varies one parameter; each column shows one site. The deviation from baseline (in parentheses) is shown for each bar; green = above baseline, red = below baseline, white = baseline. All deviations are $|\Delta P(H)| \leq 0.033$, and qualitative rankings are preserved across all 27 parameter combinations tested. Generated by `scripts/generate_sensitivity_analysis.py`; output at `outputs/diagnostics/sensitivity_analysis.pdf`.

4.7 Future Work

The three most impactful extensions to this framework are: (1) *post-Dragonfly calibration* — DraMS organic abundances and DraGNS hydrogen maps will directly validate f_2 and f_8 at Selk, enabling quantitative recalibration of weights and prior means; (2) *full multivariate posterior* — replacing the independence assumption (Section 4.5) with a full covariance structure estimated from high-resolution SAR patches where multiple features co-occur; and (3) *additional temporal anchors* — particularly the early lake-formation epoch (-2.0 to -0.5 Gya) where the liquid-hydrocarbon ramp most sensitively determines whether equatorial or polar sites lead the transition.

4.8 Conclusions

This thesis presents the first quantitative, spatially-resolved, multi-epoch Bayesian habitability map of Titan's surface, derived from eight observational proxy features from the complete Cassini dataset. Conclusions are organised by research question.

RQ1: Spatial distribution of present-epoch habitability. North polar lake margins are the most habitable present-day environments. Kraken Mare's southern shoreline achieves $P(H) \approx 0.44$ (rank 1), driven by coincident maximum liquid hydrocarbon (f_1=1.0), active methane cycle ($f_4 \approx 0.70$), and high shoreline geodiversity ($f_7 \approx 0.76$). At the present epoch all top-10 sites are lake or named lacus margins; equatorial dune fields rank 11th and below despite high organic abundance, limited by the absence of surface liquid. All eight feature-weight/importance ratios lie within 10% of unity, confirming that the geographic discrimination of each Cassini observable matches independent astrobiological reasoning.

RQ2: LHB habitability and impact structures. During the LHB, cryovolcanic features lead (Hotei Regio $P(H)$=0.295; Sotra Facula $P(H)$=0.289) because their elevated subsurface ocean proximity and surface–atmosphere coupling combine with the globally boosted acetylene proxy. Craters (Menrva $P(H) \approx 0.27$; Selk $P(H) \approx 0.26$) rank 11th and lower in the 8-feature thesis formula; their impact-melt advantage is explicitly captured in the animation model. Selk's Shangri-La dune–plains boundary maximises organic substrate for Strecker synthesis in the transient melt pond, consistent with Neish et al. (2018).

RQ3: Multi-epoch temporal trajectory. The trajectory is non-monotonic across three regimes. (i) LHB (−3.8 to −1.5 Gya): cryovolcanic and impact-melt dominance. (ii) Polar-lake (−1.0 to +4.0 Gya): lake or lacus margins rank 1–10 with stable rankings across three epochs. (iii) Global ocean (+5.1–6.0 Gya): all sites score $P(H)>0.42$; cryovolcanic sites lead as their organic substrates dissolve into the ocean. A transient astrobiological minimum at $\sim +5.0$ Gya precedes the eutectic crossing. The red-giant ocean at +5.9 Gya, sustained for ∼800 Myr over accumulated organics, represents an extraordinary future habitability scenario unmatched elsewhere in the solar system.

RQ4: Validation of the Dragonfly landing site. Selk crater has the highest geodiversity of any equatorial site (f_7=0.629, 7.8× the global median), derived from SAR data without reference to mission planning. This terrain-boundary diversity — dune, plains, labyrinth, and crater floor within 45 km — is the Cassini proxy for the organic–water-ice mixing interface targeted by Dragonfly's prebiotic chemistry instruments. Selk's lower present-epoch solvent score ($P(H)$=0.24) is scientifically consistent: Dragonfly targets *past* liquid-water chemistry, correctly identified by the framework's two-framework distinction (Section 4.1.2).

Together, these results demonstrate that Titan possesses a multilayered astrobiological significance from its geological past through the present to the far future, quantifiable from the Cassini dataset with sufficient precision to guide mission planning and interpret in-situ measurements.

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A Physical Parameters for Titan

Table A.1: Physical constants and observationally-constrained parameters for Titan used throughout this thesis.

Parameter	**Value**	**Reference**
Mean radius	2575.0 km	IAU 2015
Surface gravity	$1.352\,\mathrm{m\,s^{-2}}$	Zebker et al. (2009)
Surface pressure	1.467 ± 0.001 bar	Fulchignoni et al. (2005)
Surface temperature	93.65 ± 0.25 K	Fulchignoni et al. (2005)
N_2 fraction (surface)	$\approx 94.2\%$	Niemann et al. (2005)
CH_4 fraction (surface)	$5.65 \pm 0.18\%$	Niemann et al. (2005)
Tidal Love number k_2	0.589 ± 0.150	Iess et al. (2012)
Saturn orbital period	29.46 yr	—
Axial obliquity	26.73°	—
Solar flux at Titan	$\approx 14.9\,\mathrm{W\,m^{-2}}$	at 9.54 AU
Organic deposition rate	$\sim 5 \times 10^{-14}\,\mathrm{g\,cm^{-2}\,s^{-1}}$	Lavvas et al. (2011)
North polar sea volume	$\sim 70{,}000\,\mathrm{km^3}$	Mastrogiuseppe et al. (2019)
Water–ammonia eutectic	176 K at 32.6 wt% NH_3	Grasset and Pargamin (2005)
Subsurface ocean depth	$\sim$ 55–80 km	Tobie et al. (2005)

B Bayesian Prior Specification

Table B.1: Prior means μ_i and weights w_i for all eight habitability features at the present (Cassini) epoch. Global prior mean $\mu_0 = \sum_i w_i \mu_i = 0.331$. Concentration parameter $\kappa = 5$ is applied uniformly.

Feature	**Symbol**	w_i	μ_i	**Key ref.**
Liquid hydrocarbon	f_1	0.25	0.020	Hayes et al. (2008)
Organic abundance	f_2	0.20	0.700	Malaska et al. (2025)
Chemical energy	f_3	0.20	0.350	Strobel (2010)
Methane cycle	f_4	0.15	0.400	Mitchell and Lora (2016)
Surface–atm. interaction	f_5	0.08	0.350	Miller et al. (2021)
Topographic complexity	f_6	0.06	0.250	Corlies et al. (2017)
Geomorphological diversity	f_7	0.04	0.300	Lopes et al. (2019)
Subsurface ocean proximity	f_8	0.02	0.030	Iess et al. (2012)
Global prior mean μ_0			0.331	

C Geomorphological Organic Abundance Scores

Table C.1: Organic abundance proxy scores used for the geomorphological seamless mode of Feature 2. Grounded in published VIMS spectral characterisation of each terrain class.

Class	Name	Physical basis	Score
Dunes	Equatorial sand seas	Tholin-dominated; highest VIMS ratio (Lorenz et al., 2006)	0.82
Plains	Smooth lowlands	Organic sediment blanket (Lopes et al., 2016)	0.68
Basins	Topographic lows	Organic accumulation (Lopes et al., 2020)	0.58
Labyrinth	Dissected terrain	Spectral affinity with plains (Malaska et al., 2020)	0.55
Craters	Impact structures	Mixed water-ice and organic ejecta (Neish et al., 2018)	0.35
Mountains	Ridge terrain	Water-ice-rich crustal material (Solomonidou et al., 2018)	0.25
Lakes	Hydrocarbon seas	Liquid hydrocarbons; low solid organic content (Brown et al., 2008)	0.05

D Declared Assumptions and Known Limitations

A1. Conditional feature independence. The eight features are treated as conditionally independent given habitability H. In practice, organic abundance and chemical energy are positively correlated, as are liquid hydrocarbon and surface–atmosphere interaction. The independence assumption is conservative: correlated positive features slightly overstate the posterior, but the framework remains physically motivated and the data-derived importances are consistent with the prior weights (Section 3.4.7).

A2. Static geomorphological template. The Lopes et al. (2019) map is used for all five temporal configurations. The Late Heavy Bombardment surface had different dune distributions and crater density, and the future surface will be submerged. The present template is used as the best available proxy, with amplitude modifications applied through epoch-specific prior means and scale functions.

A3. Seamless organic abundance mode. The organic abundance feature uses terrain-class scores globally rather than VIMS spectral band ratios, because the VIMS mosaic covers only ∼50% of the globe and direct blending produced a visible seam at the coverage boundary at 180°W.

A4. Past-epoch Cassini feature contamination. The past-epoch posterior was computed using present-era Cassini SAR maps, which encode the present polar lake morphology even at $t = -3.5$ Gya. The north polar Bayesian score in the past anchor is inflated above the physically expected prior value. This is mitigated in the multi-epoch reconstruction by using the analytical Bayesian formula directly for pre-lake-formation frames (Section 2.6) and is documented in Section 4.5.

A5. Multi-epoch temporal scaling accuracy. The multi-epoch reconstruction applies scalar scale functions $s_i(t)$ to the present-epoch feature maps and re-evaluates the Bayesian posterior formula at each epoch (Section 2.6.1). This is physically motivated but is not equivalent to independent epoch-specific feature extraction. The scale functions are calibrated to reproduce known physical transitions (lake formation, solar warming, eutectic crossing) but have no observational constraint at intermediate epochs. Estimated uncertainty: $\Delta P(H) \lesssim \pm 0.05$ at non-anchor epochs.

A6. SAR coverage gap. Approximately 33% of Titan's surface lacks SAR coverage. Features 1, 3, and 8 are degraded in these regions, where geomorphological and topographic proxies substitute.

E Acronyms, Notation, and Data Source Details

E.1 Acronyms and Nomenclature

The following acronyms are used throughout this chapter and the remainder of the thesis. They are defined here for convenience and also at the point of first use.

SAR Synthetic Aperture Radar. An active microwave imaging technique in which the motion of the spacecraft is used to synthesise a very long antenna aperture, yielding fine-resolution imagery of surface texture and composition.

VIMS Visible and Infrared Mapping Spectrometer. The *Cassini* instrument that imaged Titan's surface through six near-infrared atmospheric transparency windows, providing surface composition data.

CIRS Composite Infrared Spectrometer. The *Cassini* thermal emission instrument used here to constrain the surface temperature field.

GTDR Global Titan Digital elevation model, Radar — the raw Cassini RADAR altimetry dataset archived in the Planetary Data System.

GTDE Global Titan Digital Elevation model — the globally interpolated topographic product of Corlies et al. (2017) derived from the GTDR and supplementary data sources.

DEM Digital Elevation Model. A gridded representation of surface topography.

PDS Planetary Data System. The NASA archive for planetary mission data.

HDI Highest Density Interval. A Bayesian credible interval containing the specified probability mass within the narrowest possible range.

E.2 Notation

Two quantities appear throughout this chapter and are defined here to avoid repetition. Each habitability-proxy feature f_i ($i = 1, \dots, 8$) is assigned:

- A **weight** $w_i \in [0, 1]$, where $\sum_i w_i = 1$. The weight encodes the relative astrobiological importance of feature i as a habitability indicator. A feature with a higher weight exerts a larger influence on the posterior probability of habitability, all else being equal. Weights are assigned from physical reasoning and validated against data-derived importances (Section 3.4.7).

- A **prior mean** $\mu_i \in [0, 1]$. This is the expected spatially-averaged value of feature i before any observational data are considered — a statement of our prior knowledge about how common or widespread the habitability-relevant condition is on Titan. For example, liquid hydrocarbon covers $\sim$2% of Titan's surface (Hayes et al., 2008), so its prior mean is $\mu_1 = 0.02$. The weighted sum of all prior means defines the global prior mean habitability: $\mu_0 = \sum_i w_i \mu_i = 0.331$. Note that the value $\mu_0 = 0.331$ does not appear in the prior literature as a global habitability estimate for Titan; it emerges from the component prior means in Table B.1 and should be interpreted as encoding the observation that Titan's surface is globally organic-rich whilst simultaneously recognising that liquid solvent is present at only $\sim$2% of the surface. The resulting moderate prior is intentionally revisionable: the weak concentration parameter $\kappa = 5$ ensures that the data, not the prior, determine the posterior at any pixel where observational coverage is good. Full values are tabulated in Appendix B.

This notation first appears in a practical context in Section 2.3, where the eight features are described individually. Their role in the Bayesian update is explained in Section 2.4.

E.3 Observational Data Source Details

E.3.1 Cassini RADAR Synthetic Aperture Radar Mosaic

The Cassini RADAR instrument operated in SAR mode during Titan flybys, imaging the surface at $\sim$350 m per pixel over the 127 targeted flybys from 2004 to 2017 (Janssen et al., 2016). SAR measures the normalised radar cross-section σ^0 (typically expressed in dB), which is sensitive to surface roughness, dielectric properties, and the presence of smooth or conducting materials. Low backscatter ($\sigma^0 < -20$ dB) is diagnostic of smooth surfaces (consistent with liquid hydrocarbon lakes) or dielectrically absorbing material (consistent with tholin-rich dune sand). High backscatter correlates with rough, water-ice-rich material such as mountain terrain, crater ejecta, and some cryovolcanic surfaces. The SAR mosaic covers $\sim$67% of Titan's surface and constitutes the primary data source for the liquid hydrocarbon, chemical energy, and subsurface ocean features.

E.3.2 VIMS Near-Infrared Spectral Mosaic

The Cassini Visible and Infrared Mapping Spectrometer (VIMS) observed Titan's surface through atmospheric transparency windows at 0.94, 1.08, 1.27, 1.59, 2.03, and 2.79 µm. The band ratio 1.59/1.27 µm is an established proxy for tholin (complex organic) abundance at the surface: the 1.59 µm window probes deeper into the tholin-bearing surface whilst the 1.27 µm window provides atmospheric normalisation (Barnes et al., 2007). The Seignovert et al. (2019) combined VIMS+ISS mosaic (Seignovert et al., 2019) provides this band ratio map at 2–4 km per pixel coverage, but only for the 0–180°W hemisphere ($\sim$50% of the globe). The significance of this coverage gap for the organic abundance feature is discussed in Section F.2.

E.3.3 CIRS Thermal Emission Observations

The Composite Infrared Spectrometer (CIRS) measured Titan's surface brightness temperature at far-infrared wavelengths throughout the Cassini mission. Jennings et al. (2019) derived an analytical model of the latitude-dependent surface temperature from 13 years of CIRS observations, producing the formula:

$$T(\phi, Y) = (93.53 - 0.095\,Y)\cos\Big[(\phi + 0.85 - 3.2\,Y)(0.0029 - 0.00006\,Y)\Big] \tag{E.1}$$

where ϕ is latitude in degrees and $Y = t_{\mathrm{CE}} - 2009.61$ is years from the northern spring equinox (22 May 2009). This formula reproduces the observed zonal surface temperatures to within ± 0.4 K (rms) across the full Cassini epoch and provides 100% global coverage, grounded in CIRS focal-plane-1 limb and nadir observations across the full 2004–2017 mission. The model captures only latitudinal and seasonal variation; no longitudinal inhomogeneity is represented (consistent with the absence of significant longitudinal temperature gradients in the Cassini dataset). For temporal epochs other than the present, the mean annual temperature field is held stationary: this approximation is justified because the dominant habitability driver at non-present epochs is the liquid-hydrocarbon scale function $s_1(t)$ rather than the temperature gradient, and the methane-cycle feature contributes only $w_4 = 0.15$ to the posterior. It is used to synthesise the global temperature field and its meridional gradient, which enters the methane cycle feature (Section F.4).

E.3.4 GTDE/GTDR Topographic Models

The Global Titan Digital Elevation model (GTDE), produced by Corlies et al. (2017), integrates Cassini RADAR altimetry, SARtopo stereophotogrammetry, and radial basis function interpolation to produce a globally-interpolated topographic model at ~22 km per pixel (2 pixels per degree). Coverage is approximately 90% of the globe, with the south polar region and some high-latitude northern areas relying most heavily on interpolation. The topographic model contributes to the chemical energy feature (identifying topographic lows as preferential organic accumulation sites), the topographic complexity feature (through local roughness estimation), and the surface–atmosphere interaction feature (through slope calculation).

E.3.5 Lopes (2019) Global Geomorphological Map

The global geomorphological map of Lopes et al. (2019) classifies the Cassini SAR-imaged surface into seven terrain units (plains, dunes, mountains, basins, labyrinth terrain, craters, and lakes) using SAR texture, backscatter, morphology, and VIMS spectral correlation. This map provides the terrain classification that enters the organic abundance feature (as a globally seamless gap-fill) and the geomorphological diversity feature. The shapefiles are provided in east-positive geographic coordinates and require conversion to the west-positive convention of the analysis grid.

E.3.6 Miller (2021) Global Fluvial Channel Map

Miller et al. (2021) identified and mapped Titan's global fluvial channel network from Cassini SAR data, producing a vector dataset of channel locations and widths. This network constitutes the primary mechanism for organic material transport from equatorial source regions to the polar basins, and channel density is used as one component of the surface–atmosphere interaction feature.

E.3.7 Tidal Love Number and Gravity Data

The degree-2 tidal Love number $k_2 = 0.589 \pm 0.150$ from Iess et al. (2012) provides a global constraint on interior structure, confirming the presence of a subsurface liquid layer. This scalar quantity enters as a global floor value in the subsurface ocean proximity feature.

E.3.8 Hedgepeth (2020) Global Crater Catalogue

The global catalogue of confirmed and probable impact structures on Titan by Hedgepeth et al. (2020) records crater diameters and positions for all resolved impact features in the Cassini SAR dataset. This catalogue is used exclusively in the *Past* epoch to generate the impact-melt proximity feature.

F Feature Engineering: Scientific Basis and Implementation

The eight habitability-proxy features are extracted from the Cassini data sources described in Appendix E. Each subsection gives the scientific basis, prior mean justification, implementation procedure, and missing-data backfill strategy.

Eight habitability-proxy features are extracted from the above data sources. Figure F.1 shows all eight feature maps at the present epoch, illustrating their distinct spatial patterns. Figure F.2 shows how the relative activity of each feature is modulated across geological time.

All features are normalised to the range $[0, 1]$ using a robust percentile-clipping approach:

$$\hat{f}_i = \text{clip}\left(\frac{f_i - f_{i,\text{p2}}}{f_{i,\text{p98}} - f_{i,\text{p2}}},\ 0,\ 1\right) \tag{F.1}$$

where $f_{i,\text{p2}}$ and $f_{i,\text{p98}}$ are the 2nd and 98th percentiles of finite values in the global layer, mapping the central 96% of the observed dynamic range onto $[0, 1]$ whilst preventing extreme outliers from compressing the scale. Where a feature is unavailable at a particular pixel (due to data gaps), the value is set to NaN and handled through the Bayesian missing-data imputation procedure described in Section 2.4.5.

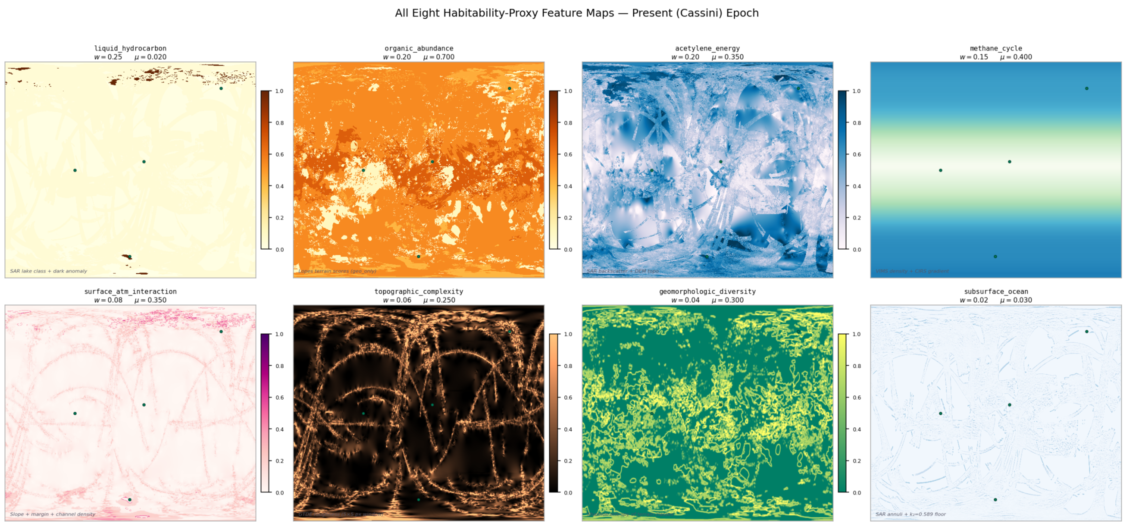

Figure F.1: All eight habitability-proxy feature maps at the present (Cassini) epoch, each normalised to $[0, 1]$. The panels illustrate the distinct spatial patterns contributed by each feature: the polar concentration of liquid hydrocarbon (f_1), the near-global organic abundance (f_2), the topographically-modulated chemical energy proxy (f_3), and the latitude-dependent methane cycle signal (f_4), amongst others. The combined posterior is derived from all eight simultaneously via the Bayesian update described in Section 2.4. Prior weight w_i and prior mean μ_i are shown for each panel.

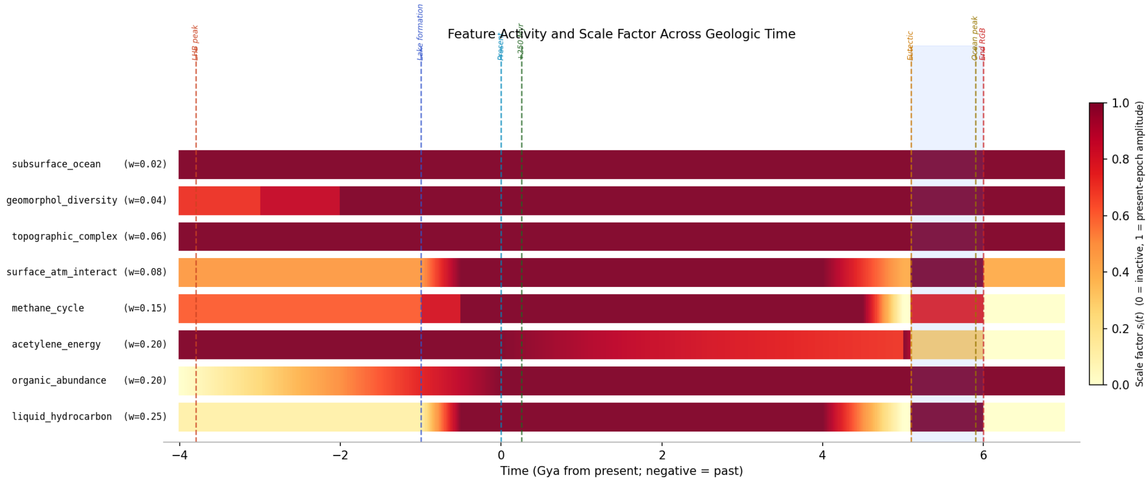

Figure F.2: Scale factor (relative activity) for each feature as a function of geological time from −4.0 Gya to +7.0 Gya. Colour intensity encodes the scale applied to each feature's prior mean at that epoch: bright yellow indicates full present-epoch amplitude; dark blue indicates near-zero activity. Key events (LHB peak, lake-formation onset, present epoch, solar warming ramp, eutectic crossing, ocean peak) are marked with vertical dashed lines. The impact-melt proxy and cryovolcanic flux features are active only in pre-lake epochs and are shown in the lower two rows.

F.1 Feature 1: Liquid Hydrocarbon Availability ($w_1 = 0.25, \mu_1 = 0.02$)

Scientific basis. Liquid methane and ethane are the proposed non-aqueous biochemical solvents for any life operating on Titan's present-day surface (McKay, 2016; McKay and Smith, 2005). At the ambient surface temperature of ∼94 K, water is structurally as rigid as rock and cannot support molecular diffusion processes fundamental to metabolism. Liquid methane and ethane are, however, stable at 94 K and 1.47 bar, as confirmed by the permanent north polar seas. Benner et al. (2004) proposed that non-polar solvents can in principle support biochemistry through different but self-consistent molecular architectures, and Stevenson et al. (2015) demonstrated through molecular dynamics simulations that acrylonitrile molecules present in Titan's atmosphere can form stable vesicle membranes in liquid methane. The presence of persistent liquid hydrocarbon is therefore treated as the single most important habitability prerequisite for surface life at the present epoch, and accordingly receives the highest prior weight ($w_1 = 0.25$).

Prior mean. The north polar seas cover approximately 1.5–2% of Titan's surface (Hayes et al., 2008), giving a prior mean of $\mu_1 = 0.02$. Despite the high weight, the low prior mean means that most pixels contribute negligibly through this feature; its habitability impact is strongly localised to the polar regions.

Implementation. The primary source is the Lopes et al. (2019) lake polygon class (terrain class 7), rasterised to the canonical grid as a binary lake mask. A secondary contribution comes from SAR backscatter dark anomalies not otherwise attributed to organic material. Confirmed empty lake basins from the Birch et al. (2017) analysis are explicitly assigned

$f_1 = 0$ to reflect the absence of present-day liquid even within geomorphologically defined basins. In the absence of SAR coverage, ISS 938 nm albedo imagery provides a proxy, with smooth, low-albedo regions consistent with liquid surfaces elevated towards higher feature values.

Backfill. In regions lacking both SAR and ISS coverage, the feature is assigned NaN and imputed at the prior mean during inference.

F.2 Feature 2: Organic Abundance ($w_2 = 0.20$, $\mu_2 = 0.70$)

Scientific basis. Tholins — the complex refractory macromolecular organic compounds produced continuously by ultraviolet photolysis of the N_2/CH_4 mixture in the upper atmosphere — are the dominant surface material across most of Titan. Their astrobiological significance is multifaceted. Laboratory hydrolysis of tholins in liquid water at biologically relevant temperatures (273–373 K) produces amino acids, nucleobase analogues, and sugar-like polyols at yields of $\sim$0.1% by mass (Neish et al., 2018), directly confirming that tholins can serve as a chemical feedstock for prebiotic synthesis when liquid water is available — either from impact melts at past epochs or from the red-giant ocean in the far future. Meyer-Dombard et al. (2025) characterise the tholin layer as the primary chemical substrate for any surface habitability scenario on Titan. The feature weight ($w_2 = 0.20$) is second only to liquid hydrocarbon, reflecting this central role.

Prior mean. The revised prior mean of $\mu_2 = 0.70$ (updated from 0.60 in earlier pipeline versions) reflects updated spectral characterisation from Malaska et al. (2025) and Cable et al. (2012), indicating that approximately 82% of the SAR-classified surface by area falls into organic-rich terrain classes (dune: 17%, plains: 65%). The organic abundance feature consequently makes the largest single contribution to the global prior mean ($w_2\mu_2 = 0.140$).

Implementation and the seamless organic mode. The primary spectral data source is the VIMS+ISS 1.59/1.27 µm band-ratio mosaic from Seignovert et al. (2019), but this covers only the 0–180°W hemisphere ($\sim$50% of the globe). Extensive testing of direct blending between the VIMS band ratios and geomorphological class scores in the 180–360°W hemisphere produced a visible seam artefact at the 180°W boundary: the absolute calibration of the VIMS spectral ratio differs systematically from the terrain-class score at the coverage boundary, producing a step change of $\sim$0.13 in median organic abundance clearly visible in rendered maps. This artefact cannot be eliminated by a single global offset because the boundary region terrain is not representative of all terrain classes. The analysis therefore uses a seamless geomorphological mode, applying the terrain-class scores from Table C.1 (Appendix C) globally. Whilst this approach sacrifices per-pixel spectral variation within the VIMS-covered hemisphere (reducing the standard deviation from 0.27 for VIMS data to 0.16 for terrain-class scores), it ensures that no spatial artefact from the uneven VIMS coverage contaminates the habitability maps.

Backfill. The Lopes (2019) terrain classification covers ~90% of the SAR-imaged surface (~67% of the globe). In regions outside SAR coverage, terrain class is estimated from ISS albedo units with reduced specificity.

F.3 Feature 3: Chemical Energy Availability ($w_3 = 0.20$, $\mu_3 = 0.35$)

Scientific basis. Life requires a sustained source of chemical free energy to drive metabolic reactions against thermodynamic equilibrium. On Titan's surface, the primary candidate is acetylenotrophy ($C_2H_2 + 3\,H_2 \longrightarrow 2\,CH_4$, $\Delta G = -334\,\mathrm{kJ\,mol^{-1}}$; where ΔG is the Gibbs Free Energy change of the reaction, a negative value indicating spontaneous energy release) available from acetylene–hydrogen combination (McKay and Smith, 2005). Yanez et al. (2024) recalculated the available energy under updated thermochemical conditions at Titan's surface as 69–78 $\mathrm{kJ\,mol^{-1}}$ C, well above the minimum for terrestrial methanogens. Acetylene is produced continuously by UV photolysis of methane and sediments to the surface where it has been directly detected by GCMS (Niemann et al., 2005). The chemical energy feature is therefore a proxy for the co-availability of acetylene and hydrogen at the surface.

Implementation. Since the acetylene surface abundance cannot be directly mapped by any Cassini instrument, it is approximated through two components. SAR backscatter (weight 70% where coverage exists): SAR-dark surfaces characteristic of organic-rich terrain represent regions where acetylene is most likely to accumulate without competing abiotic reactions consuming it rapidly. Topographic inversion (weight 30%): acetylene and other atmospheric condensates preferentially accumulate in topographic lows through both gravitational settling and fluvial transport; the normalised negative of the GTDE elevation ($-z$) assigns high values to topographic lows. In SAR-absent regions, the feature reverts to the pure topographic inversion component.

F.4 Feature 4: Methane Cycle Intensity ($w_4 = 0.15$, $\mu_4 = 0.40$)

Scientific basis. The methane cycle is the dominant agent of chemical transport on Titan. It delivers organic material from equatorial photochemical production regions to the polar lake margins, maintains chemical gradients at shorelines, and provides seasonal wet–dry cycling at lake edges that some prebiotic chemistry scenarios identify as important for concentrating organic reactants and driving condensation reactions (Benner et al., 2004; Mitchell and Lora, 2016). Regions experiencing the most active methane cycling receive fresh organic inputs most regularly and maintain the most dynamic chemical environments.

Implementation. The methane cycle intensity is a weighted composite of three independently computed components:

1. **VIMS observation density** (50%): the number of VIMS spectral footprints per grid cell, smoothed with a 5-pixel Gaussian kernel. Cassini's VIMS observation tar-

geting was scientifically motivated, prioritising regions of active methane-cycle interest; higher coverage density therefore correlates with active methane cycling zones (Le Mouélic et al., 2019).

2. **Meridional temperature gradient** $|\partial T/\partial \phi|$ (25%): derived from the Jennings (2019) model (Eq. E.1), evaluated at $Y = +1.39$ (year 2011.0, the approximate mission midpoint). Regions where the surface temperature gradient is steepest experience the strongest differential evaporation and precipitation, driving the most vigorous methane transport.

3. **Latitude prior** (25%): the function $\exp[-(\phi \pm 45)^2/(25)^2]$, following Mitchell and Lora (2016)'s general circulation modelling result that the mid-latitude regions host the primary methane cycle activity during the Cassini epoch.

This feature has 100% global coverage by construction and requires no backfill.

F.5 Feature 5: Surface–Atmosphere Interaction Intensity ($w_5 = 0.08$, $\mu_5 = 0.35$)

Scientific basis. The physical interface between surface and atmosphere is where gas-phase photochemical products encounter potential liquid solvents and organic substrates. Lake margins and fluvial channel termini are the primary loci of this exchange. Hayes (2016) describes the complex sedimentological and chemical processes at hydrocarbon lake shorelines, including the evaporation-driven concentration of dissolved organics and the seasonal flooding and drying cycles that expose organic sediments to liquid contact. Miller et al. (2021) documents that the fluvial channel network terminates preferentially at polar lake margins, establishing these as zones of maximum organic delivery and chemical gradient activity.

Implementation. A weighted sum of three spatial components:

$$f_5 = 0.30\,\hat{s} + 0.40\,\hat{m} + 0.30\,\hat{c} \tag{F.2}$$

where $\hat{s}$ is the normalised topographic slope $|\nabla z|$ from the GTDE DEM (high slopes drive physical transport and mixing); $\hat{m}$ is a binary lake-margin mask extended to $\sim$45 km from the nearest confirmed lake shore; and $\hat{c}$ is the Gaussian-smoothed ($\sigma = 3$ pixels) channel density from the Miller et al. (2021) dataset, normalised at the 2nd–98th percentile range. The lake-margin component receives the highest within-feature weight (40%) because shorelines concentrate all three relevant processes — liquid contact, organic delivery, and thermal/chemical cycling.

F.6 Feature 6: Topographic Complexity ($w_6 = 0.06$, $\mu_6 = 0.25$)

Scientific basis. Local topographic roughness at $\sim$20 km scale correlates with the dissected terrain types that most frequently expose water-ice-bearing crustal material (Lopes et al., 2019). On Earth, topographic heterogeneity at ecologically relevant scales creates micro-environmental diversity — variations in drainage, aspect, moisture, and chemical exposure — that underpins elevated species richness at landscape boundaries. On Titan, the analogous principle is that varied topography produces diverse chemical micro-environments with potentially richer reactive diversity.

Implementation. Local standard deviation of GTDE elevation computed in a 5×5 pixel ($\sim$22 km) sliding window, normalised to $[0, 1]$ at the 2nd–98th percentile range. South polar gaps in the primary GTDE are filled with the Corlies et al. (2017) globally interpolated DEM prior to roughness computation.

F.7 Feature 7: Geomorphological Diversity ($w_7 = 0.04$, $\mu_7 = 0.30$)

Scientific basis. Astrobiologically relevant chemistry is disproportionately likely to occur at interfaces between geochemically distinct terrain types. The reasoning is thermodynamic: chemical gradients — differences in redox potential, organic content, water-ice availability, or pH — drive reactions that would not proceed in chemically uniform settings (Shock and Holland, 2007). On Titan, the most important geochemical interfaces identified from Cassini data are:

1. The **dune–plains boundary**, where organic-rich tholin-saturated dune sand is in direct lateral contact with the moderate-organic plains through which fluvial channels transport material towards the polar lakes. This interface concentrates the exchange between the primary photochemical organic source (dunes) and the active transport system (plains channels).

2. The **crater ejecta interface**, where water-ice-rich crustal material excavated by impact is juxtaposed with adjacent organic-bearing terrain. Neish et al. (2018) demonstrated experimentally that mixing water ice and tholins under liquid water conditions produces amino acids; the spatial co-occurrence of these two materials at crater ejecta boundaries is therefore a direct proxy for the relevant chemical reactivity.

3. The **lake–plains/dune boundary**, where liquid hydrocarbon meets the organic substrate, enabling dissolution and transport of surface organics into the lake system.

The Shannon entropy of terrain class distribution within a local window quantifies the degree of terrain-type mixing at each pixel. A pixel surrounded by a single terrain class (entropy $= 0$) sits in a geochemically homogeneous environment. A pixel at the intersection of four terrain classes (entropy approaching H_{max}) sits at a multi-way geochemical interface. This feature receives the lowest weight of the present-epoch set ($w_7 = 0.04$) because it is an indirect and geometrical proxy for chemical gradients rather than a direct measurement.

Implementation. The Shannon entropy of the terrain class distribution in a sliding window of radius 10 pixels (~45 km), normalised by the theoretical maximum entropy for the number of classes present:

$$f_7(r,c) = \frac{H(r,c)}{H_{\max}}, \qquad H(r,c) = -\sum_{k=1}^{N_c} p_k(r,c) \ln p_k(r,c), \qquad H_{\max} = \ln N_c \tag{F.3}$$

where $p_k(r,c)$ is the fraction of window pixels classified as terrain class k, and N_c is the number of distinct classes present in the window. Shannon entropy is the standard information-theoretic measure of diversity (Shannon, 1948) and has been applied to quantify geomorphological and ecological diversity in planetary and terrestrial settings (Lorenz, 2014). The normalisation by $H_{\max} = \ln N_c$ maps the maximum possible entropy (uniform distribution across all present classes) to 1.0, ensuring the feature is comparable across pixels with different numbers of terrain classes.

F.8 Feature 8: Subsurface Ocean Proximity ($w_8 = 0.02$, $\mu_8 = 0.03$)

Scientific basis. Titan's confirmed subsurface water–ammonia ocean (Iess et al., 2012) is potentially habitable by conventional criteria (Affholder et al., 2025), and there is indirect evidence that material from this ocean has reached the surface at specific locations. SAR-bright annular features surrounding several impact craters are most consistently interpreted as ancient impact-melt or cryolava fronts where liquid spread outward from basin centres, potentially establishing chemical contact between the subsurface ocean and the organic surface inventory (Wood et al., 2010). The prior mean is set at $\mu_8 = 0.03$ (revised downward from 0.10 following Neish et al. 2024), reflecting the genuine uncertainty in the surface expression of the subsurface ocean at the present epoch.

Implementation. The feature is constructed in two parts. First, a global floor value of $p_{\text{floor}} = 0.03$ is applied uniformly to all pixels, reflecting the global presence of the subsurface ocean as confirmed by the tidal Love number k_2 (Iess et al., 2012). This floor captures the fact that the ocean is a global structure and that the possibility of ocean–surface exchange cannot be excluded at any location, even if no surface expression is visible.

Second, localised enhancements above the floor are applied at SAR-bright annular structures identified in the Cassini RADAR mosaic. These annuli are selected by their backscatter ($\sigma^0 > 0$ dB, indicating rough or water-ice-rich material), their plan-view morphology (roughly circular or lobate, distinct from the linear patterns of mountain ridges or the radar-dark texture of dunes), and their association with confirmed or probable impact structures in the Hedgepeth et al. (2020) catalogue. At each such structure, a two-dimensional Gaussian kernel of half-width $\sigma \approx 200$ km centred on the crater creates a smooth spatial enhancement. The 200 km scale is set to approximate the spatial extent of the ancient melt-water or cryolava deposit that the annulus represents: for a 90 km diameter crater (Selk), melt-front runout distances of ~200–500 km are consistent with the low-viscosity water–ammonia melt and the low-gravity environment (O'Brien et al., 2005).

The feature is the normalised sum of the global floor and all Gaussian kernels, clipped to $[0, 1]$. Because the global floor is already low ($\mu_8 = 0.03$) and only a small number of confirmed annuli exist in the SAR mosaic, the spatial variance of f_8 is very low: it is 0.03 everywhere except within ∼500 km of a confirmed annular structure, where it reaches values of 0.15–0.25.

G The Cassini–Huygens Mission and Dragonfly

The *Cassini–Huygens* mission (2004–2017) provided the observational foundation of this thesis through 127 targeted Titan flybys, the *Huygens* atmospheric probe descent and landing, and a systematic multi-instrument survey of Titan's surface, atmosphere, and interior. SAR mapping covered ∼67% of the surface; VIMS near-infrared spectroscopy ∼50%; CIRS thermal mapping the full globe at ∼10° resolution. The mission's key unresolved gap is the absence of a quantitative, spatially-resolved, multi-epoch habitability analysis integrating all data modalities — which this thesis provides.

NASA's *Dragonfly* rotorcraft (Lorenz et al., 2018), due to arrive in 2034, will traverse ∼175 km from the Belet dune field to Selk crater over a 2.7-year mission. Chapter 3 provides a quantitative evaluation of that site selection relative to globally optimal alternatives.